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ONTARIO GEOLOGICAL SURVEY

Open File Report 5447

Ontario Geoscience Research Grant Program Grant No. 5 - Part I. Component Magnetization of Algoman Banded Formations and Deposits in Ontario

by

D.T.A. Symons and M. Stupavsky

Part II. Magnetization Characteristics of the Algoman Iron Formation and Host Rocks at the Griffith Mine, Red Lake, Ontario

by

D.T.A. Symons, I.A. Osmani and M. Stupavsky

1983

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Ontario Geological Survey

ONTARIO GEOSCIENCE RESEARCH GRANT PROGRAM

Final Research Reports, 1983

Preface

This publication includes one final report on a research project that terminated March 31, 1981 and was funded under the Ontario Geoscience Research Grant Program. A requirement of the Program is that recipients of grants are to submit final reports within six months after termination of funding.

A final report is defined as a comprehensive summary stating the findings obtained during the tenure of the grant, together with supporting data. It may consist, in part, of reprints or preprints of publications and copies of addresses given at scientific meetings.

It is not the intent of the Ontario Geological Survey to formally publish the final reports for wide distribution but rather to encourage the recipients of grants to seek publication in appropriate scientific journals whenever possible. The Survey, however, also has an obligation to ensure that the results of the research are made available to the public at an early date. Although final reports are the property of the applicants and the sponsoring agencies, they may also be placed on an open file. This report is intended to meet this obligation.

E.G. Pye
Director
Ontario Geological Survey

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Grant No. 5 -

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by

D.T.A. Symons and M. Stupavsky[1]

Part II. Magnetization Characteristics of the Algoman Iron Formation and Host Rocks at the Griffith Mine, Red Lake, Ontario

by

D.T.A. Symons, I.A. Osmani[1] and M. Stupavsky

[1]Department of Geology, University of Windsor
Windsor, Ontario N9B 3P4

Manuscript approved by E.G. Pye, Director, Ontario Geological Survey, April 20, 1983.

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I. COMPONENT MAGNETIZATION OF ALGOMAN BANDED FORMATIONS AND DEPOSITS IN ONTARIO

by

D.T.A. Symons and M. Stupavsky

Department of Geology, University of Windsor

Windsor, Ontario, N9B 3P4

INTRODUCTION

This study was performed as four related but separate studies of Algoman-type banded iron formations (BIF) in Ontario. The results of the study for the Sherman Mine near Temagami have been reported in full by Symons and Stupavsky (1979, 1980), for the Moose Mountain Mine near Capreol by Symons et al (1980), for the Adams Mine near Kirkland Lake by Symons et al (1981), and for the Griffiths Mine near Red Lake by Symons et al (1982). The latter study forms the second article in this Open File Report. The purpose of this article is to summarize the results of the four studies primarily in tabular form with relevant comments. Details concerning the geology, sampling methods and locations, experimental and statistical methods, and detailed interpretations are given in the original references.

GEOLOGICAL FEATURES

All of the four iron formation deposits are Archean BIF of the magnetite oxide facies with three located in the Abitibi Belt and one in the Uchi Belt of the Superior Province in the Canadian Shield. The BIF horizons range in thickness from ~50 to ~150 m and are interlayered with siliceous metasediments associated with Archean metavolcanics of basic to acidic composition. These strata were deformed during the Kenoran Orogeny at ~2.55 Ga at which time minor basic and large acidic plutons were emplaced adjacent to or within a few kilometers of the deposits. All were isoclinally deformed to produce intermediate to steeply dipping limbs.

Table 1. Summary of Geologic Features

Feature	Sherman Mine	Moose Mountain Mine	Adams Mine	Griffiths Mine
Belt of Superior Province	Abitibi	Abitibi	Abitibi	Uchi
Host Rocks				
- Age	Archean	Archean	Archean	Archean
- Host	metasandstone and metagreywacke	metasedimentary schists & tuffs	metasedimentary tuffs & quartzites	metasedimentary greywackes
- Association	mafic to felsic metavolcanics	mafic & felsic metavolcanics	mafic to felsic metavolcanics	mafic to felsic metavolcanics
Structure				
- Folding	isoclinal, two parallel limbs	isoclinal, 11 fault offset lenses	isoclinal, syncline	isoclinal, double syncline
- Strike of Limbs	~N70°E	N0°E to N70°W	N30°E to N90°E	N30°E to N30°W
- Dip of Limbs	steep to NNW	steep to NE	intermediate to steep SE	steep to W
- Plunge	low angle	steep to N	~60°SW	~30°SW to 60°SE
Metamorphism				
- Regional	low greenschist	middle greenschist	middle greenschist	middle greenschist
- Contact	negligible	common, epidote maximum	reaches garnet amphibolite near Lebel Stock	reaches upper amphibolite near Bruce Pluton
Iron Formation				
- Type	magnetite oxide facies	magnetite oxide facies	magnetite oxide facies	magnetite oxide facies
- % soluble Fe	25	29	22	27
- Single Limb Thickness	100 m	50 m	150 m	60 m
- Specific Gravity Average for Ore (SG)	3.76	3.93	3.50	3.60
Sampling				
- Host Rock Sites	71	37	45	54
- Host Rock Specimens	~700	492	~450	540
- IF Block Samples	204	115	171	175
- IF Specimens	~805	706	~1000	1070

Repetition of the BIF member in adjacent faulted segments or in synclinal structures provided more economic thicknesses of ore at the Adams and Griffiths Mines. Subsequently minor relatively undeformed late dikes of generally basic composition were intruded into the Archean rocks. In the Abitibi Belt, these rocks are unconformably overlain by Huronian strata of ~2.3 Ga age. All of the BIF have been regionally metamorphosed to the lower to middle greenschist facies, however, significant portions of all except the Sherman Mine have been contact metamorphosed by adjacent plutons into the amphibolite facies.

Sampling was intensive. The host rocks (HR) are represented by an average of 54 sites (~546 specimens) per deposit, and the BIF are represented by an average of 166 block samples (~895 specimens) per deposit. The quoted averages for the magnetic characteristics of each deposit therefore represent the value derived from several hundreds of measurements in each case.

MAGNETIC CHARACTERISTICS

The low field magnetic susceptibility values perpendicular to banding for the BIF ($k_{\perp IF}$) fall in a narrow range, and average 6.1×10^{-2} cgs cm^{-3} (Table 2). The values for HR (k_{HR}) have a much broader range reflecting the variety of rock types sampled at each mine, however, because the ratio of $k_{\perp IF}/k_{HR}$ is >100 in all four cases it is evident that the HR do not contribute significantly to the measured magnetic anomaly over the BIF. The average susceptibility of economic ore for the BIF deposits is $10.5 \pm 3.6 \times 10^{6}$ (standard deviation) cgs cm^{-3}. Also the average equation for relating

Table 2. Summary of Rock Magnetic Characteristics

Characteristic	Sherman Mine	Moose Mountain Mine	Adams Mine	Griffiths Mine
Low Field Magnetic Susceptibility				
- $k_{\perp IF}$ cgs cm^{-3}	9.6×10^{-2}	6.2×10^{-2}	4.5×10^{-2}	4.0×10^{-2}
- k_{HR} cgs cm^{-3}	4.9×10^{-5}	5.0×10^{-5}	2.4×10^{-4}	3.6 x 10-4
- $k_{\perp IF}/k_{HR}$	188	1200	192	111
- relation $k_{\perp IF}$ & S.G. & corelation co-efficient	$k_{\perp IF}$=0.091SG-0.25 +0.84	$k_{\perp IF}$=0.091SG-0.25 +0.84	$k_{\perp IF}$=0.070SG-0.18 +0.73	$k_{\perp IF}$=0.116SG-0.31 +0.85
- - $k_{\perp IF}$ ore cgs cm^{-3}	9.6×10^{-2}	11.0×10^{-2}	$6.45 + 10^{-2}$	15.0×10^{-2}
Anisotropy of Magnetic Susceptibility				
- $k_{\parallel IF}/k_{\perp IF}$ for small core	1.60	1.65	1.55	1.45
- $k_{\parallel IF}/k_{\perp IF}$ for large core	1.70	1.75	1.69	(1.59)
-$k_{\parallel IF}/k_{\perp IF}$ accepted	1.65	1.70	1.62	1.52
Natural Remanent Magnetization				
- NRM_{IF} emu cm^{-3}	2.5×10^{-2}	1.1×10^{-2}	1.2×10^{-2}	2.1×10^{-2}
- NRM_{HR} emu cm^{-3}	2.9×10^{-6}	2.0×10^{-6}	6.0×10^{-6}	14.7×10^{-6}
- Q_{IF}, Koenigsberger ratio	0.46	0.63	0.67	0.66
- Q_{HR}, Koenigsberger ratio	0.104	0.063	0.013	0.051
- net augmenting factor	0.22	0.24	0.17	0.14

specific gravity (SG) of the BIF to susceptibility is $k_{\perp IF} = 0.092\ SG - 0.25$.

BIF has a very strong layer-controlled anisotropy of magnetic susceptibility (AMS) such that the susceptibility parallel to banding ($k_{\parallel IF}$) is about 1.62 times greater than that perpendicular to banding ($k_{\perp IF}$), meaning that the measured magnetic response over the BIF is greatly enhanced relative to the expected value if the BIF is aligned nearly parallel to the Earth's magnetic field (EMF) or greatly diminished if perpendicular to the EMF.

The measured natural remanent magnetization (NRM) for BIF (NRM_{IF}) in these deposits fall within a consistent narrow range of $1.7 \pm 0.7 \times 10^{-2}$ emu cm^{-3}. Their Koenigsberger ratio of remanent to induced magnetization (Q_{IF}) also fall within a narrow range of 0.61 ± 0.10. The values for the HR (NRM_{HR}) have an even broader range $6.4 \pm 5.8 \times 10^{-6}$ emu cm^{-3} consistent with the mixture of lithologies involved. Similarly Q_{HR} has a broader range of 0.058 ± 0.37. The important point coming out of these measurements is that the NRM_{IF} is a significant contributor to the measured magnetic anomaly whereas the NRM_{HR} is so weak that it can be neglected. The net augmenting factor (NAF) is a measure of the increase in the measured field caused by the NRM component. For the four deposits NAF averages 0.19 ± 0.05 which means that the induced magnetization component of a BIF is increased by ~19% by the addition of the NRM components.

MAGNETIC ANOMALY CALCULATION

The conventional interpretation of a magnetic anomaly utilizes the following parameters:

1) a specific geometric model for the anomaly source. BIF deposits are chemical sediments which were originally deposited with long horizontal extents and relatively narrow thickness. Thus they are usually interpreted using a semi-infinite tabular sheet model using equations for a thin or thick sheet. We use the thick sheet model following equations from Telford et al (1976);

2) the known average regional Earth's magnetic field (EMF) values for its declination, inclination, and intensity at the BIF location;

3) magnetometer data including the measured component (i.e., the total field, the vertical field, etc.), the terrain clearance of the magnetometer, and the residual anomaly characteristics (i.e., the difference between the measured magnetization of the anomaly and the background magnetization if the anomaly were not present);

4) available geologic constraints which may be present for the BIF. These include: the length and strike direction which may be known from mapping but which also may be reliably estimated from the strike length magnetic profile and the elongation direction of the anomaly; the down-dip extent which is generally unknown; the dip angle which may or may not be known from mapping or drilling; the thickness of the BIF which may or may not be constrained by mapping or drilling; the low field magnetic susceptibility which may also be constrained if outcrop samples or core are available for measuring; and the thickness of overburden which may be zero if the BIF outcrops or may be known from drill data. Usually few, if any, of these constraints are known, so that the objective of analyzing the anomaly is to put realistic values on them. The most important

parameters that are generally desired are estimates for the dip, thickness and percentage magnetite. With BIF of economic interest there are three factors which are generally not included in the calculation and thereby render a conventional interpretation meaningless. These are the AMS, the demagnetizing factor, and the NRM in order of relative importance.

Anisotropy of Magnetic Susceptibility

Noting that the average susceptibility for economic ore is $k_{\perp IF} = 10.5 \times 10^{-2}$ cgs cm^{-3}, and that $k_{\parallel IF} = 1.62\ k_{\perp IF}$, the average bulk magnetic susceptibility (BMS) for economic BIF ore of the four deposits is ~14.5×10^{-2} cgs cm^{-3}. This average corresponds to the expected value when the iron formation banding is intersected at an angle of ~32° to the EMF direction. The induced magnetization response is enhanced by +17% when the EMF is parallel to the banding and diminished by -25% when the EMF is perpendicular to the banding. Comparison of the anisotropy ratios ($k_{\parallel IF}/k_{\perp IF}$) for the four deposits with respect to metamorphic grade reveals little difference. For example the average of the two ratio values for the less highly metamorphosed Sherman and Adams Mines is the same as the average for the two more highly metamorphosed Moose Mountain and Griffiths Mines, i.e., 1.63 to 1.61.

Demagnetizing Factor

The effect of the demagnetizing factor (DF) has been examined by Gay (1963). The value ranges from 0 for a long needle of susceptible material that is parallel to the EMF to 4π for a needle perpendicular to the EMF. For a thick tabular sheet the value is 2π provides a good estimate. Thus when BIF layers are

parallel to the EMF, the DF has no effect; but when the EMF is perpendicular to the BIF layers, the measured response will be reduced to 68% of the expected value.

Natural Remanent Magnetization

The NAF for the NRM of the four deposits is 0.19 or 19%. Given the large variety of attitudes of the BIF in these deposits, the NAF represents a reasonable value regardless of the attitude of a given BIF. The NAF represents the swing of viscous remanence components (VRM) into alignment with the EMF, and thus has the effect of increasing the induced magnetization regardless of the attitude of the BIF. The presence of significant VRM components is well documented from alternating field (AF) demagnetization, thermal (Th) demagnetization, and VRM decay tests such as storage and shock tests. The hard remanence components cannot be calculated in general terms unless the specific orientation of their vectors and their magnitudes are known.

Summary

Table 3 provides a general summary of the interaction for the AMS, DF, and NRM effects in BIF. The result is illustrated in Figure 1. Stated simply, the peak response of BIF that is parallel to the EMF in northern Ontario will be ~36% greater than expected using conventional anomaly interpretation formulas whereas if the BIF is perpendicular to the EMF it will be ~30% smaller than expected.

The effect of these factors on the shape of the anomaly is to remove virtually all dip-induced skew of the anomaly. Thus in northern

Table 3. Summary of the Change in Peak Amplitude Expected for Typical BIF Ore Deposits Relative to the Conventionally Calculated Peak Amplitude

Factor	EMF $\perp$ to BIF Layers	EMF @ 32° to BIF Layers	EMF // to BIF Layers
AMS	0.75	1.00	1.17
DF	0.68	0.95	1.00
AMSxDF	0.51	0.95	1.17
NRM	+0.19	+0.19	+0.19
All Three Factors	0.70	1.14	1.36

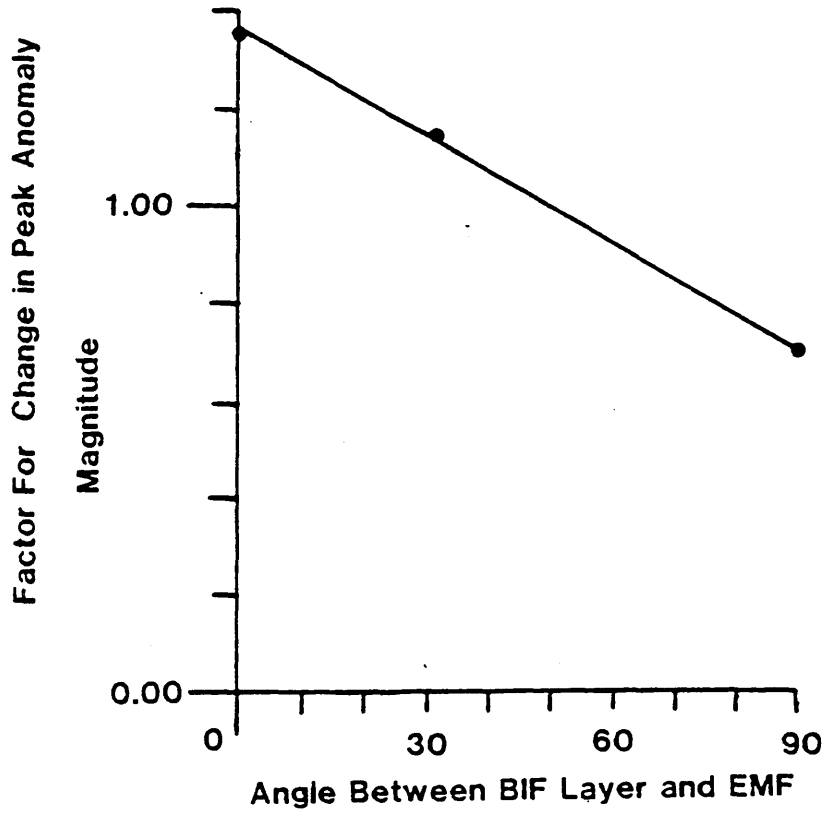


Figure 1. Relationship between the angle at which the Earth's magnetic field intersects the banding layers and the change in the magnetic anomaly peak magnitude caused by the anisotropy of magnetic susceptibility, demagnetizing factor, and natural remanent magnetization for banded iron formations of economic interest in northern Ontario.

Ontario, the shape of the measured anomaly will be nearly symmetrical. When the dip of the BIF is then calculated in a conventional way, it will be in the range of ~50°N to vertical regardless of the true dip.

From the above discussion it is evident that exploration personnel looking for BIF in northern Ontario will typically choose only those anomalies for detailed work that are closely aligned to the EMF for which they will overestimate the "iron content thickness" product by ~36%. Conversely anomalies caused by BIF oriented at a steep angle to the EMF will be interpreted as narrow or lean steeply-dipping BIF, and hence omitted from further exploration.

PALEOMAGNETIC RESULTS

Archean paleomagnetic remanence components were isolated in all four deposits both for the BIF and the HR. Some of the deposits retain minor post-Archean components associated with late basic dike intrusion, however, these are limited in sampling extent and are not considered here. The Archean pole positions are listed in Table 4 and illustrated in Figure 2 relative to the apparent polar wander path (APWP) for the Archean of North America (Seguin et al 1982). Two important points should be made. One is that fold and contact tests were used within each deposit to identify prefolding or postfolding Archean components. The second is that each BIF deposit has radically different strikes and dips, so that the agreement of the pole positions between deposits provides a very powerful fold test on a regional scale to further support the interpretation that the components are prefolding or postfolding in origin.

Table 4. Summary of Archean Pole Positions

Component	Tectonic Correction	Pole Position				Carrier Units
		Long. °W	Lat. °N	δ_p °	δ_m °	
Sherman Mine						
SM1	Cor.	352	-6	5	9	IF > HR
SM2	Cor.	260	43	4	9	IF & HR
Moose Mountain Mine						
MM1	Cor.	344	8	4	8	IF & HR
MM2	Cor.	253	44	4	8	IF & HR
Adams Mine						
AM1	Cor.	338	6	2	3	IF
AM2	Unc.	112	56	8	8	HR
Griffiths Mine						
GM1	Cor.	30	28	5	6	IF & HR
GM2	Cor.	304	14	2	4	IF > HR
GM3	pCor.	291	53	4	7	IF & HR

Note: Components are relabeled from the original papers to place them in chronologic order. AM1 is listed in error as 37.9°W. longitude in original instead of 337.9°W. The tectonic correction is given as corrected (Cor.), partially corrected (pCor.) or uncorrected (Unc.).

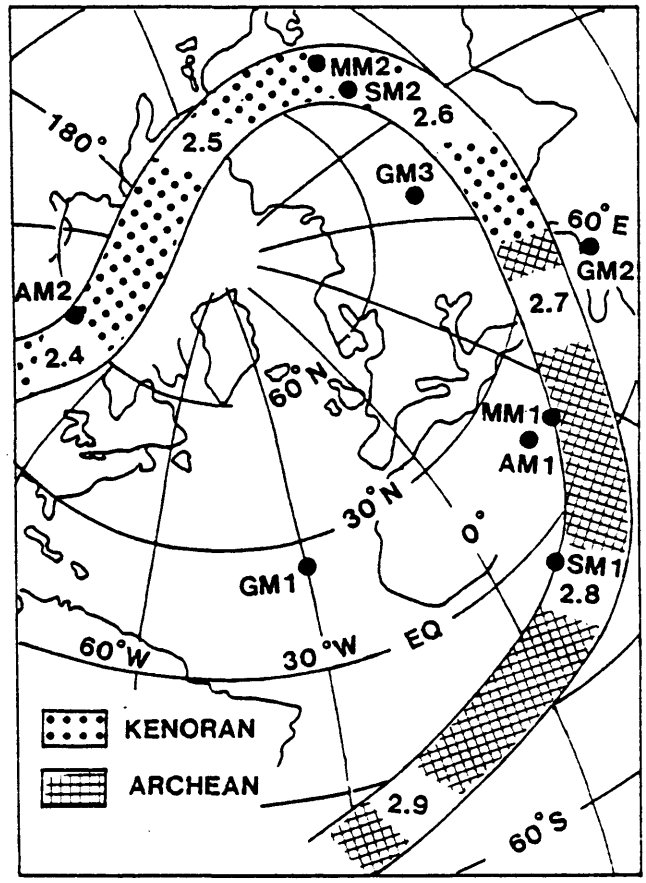


Figure 2. Apparent polar wander path for the Archean of North America (Seguin et al 1982) with the pole positions for the banded iron formation deposits (Table 4).

The poorest agreement between the APWP and the determined pole positions is for the GM1 component (i.e., Griffith Mine #1). Several explanations are noted in the paper (Symons et al 1982) which focus on the fact that this deposit is in Uchi Belt rather than the Abitibi Belt from which the majority of the APWP data have been derived. Two of the likelier explanations are the possibility that the BIF at the Griffith Mine is distinctly older than those of the Abitibi Belt so that its primary component predates the well defined portion of the Archean APWP, or alternatively that differential motions occurred between the Belts. Conversely the prefolding SM1, AM1 and MM1 components from the Abitibi Belt form a well defined cluster at ~2.75 Ga. These are considered to be primary depositional remanence components that have survived metamorphism.

Metamorphic or secondary remanence components that predate the Kenoran deformation include the SM2, MM2, and GM2. These form a second group of poles at ~2.6 Ga and are thought to relate to the onset of the Kenoran orogeny. The GM3 component could also be considered in this group because it was acquired during deformation.

Only the AM2 component is considered to be a postfolding Kenoran orogenic overprint, and it was acquired at ~2.45 Ga during cooling. Similar components are likely present in the lower thermal and alternating-field coercivity spectra of the other deposits, however, they are hard to identify with certainty because they have the same direction as VRM components associated with the EMF.

REFERENCES

Gay, S. P. 1963. Standard curves for interpretation of magnetic anomalies over long tabular ore bodies; Geophysics, Vol. 28, p. 161-200.

Seguin, M. K., Sharma, K.N.M., Woussen, G., St. Seymour, K., and Symons, D.T.A.,1982. Paleomagnetism of Archean volcanics and Proterozoic dikes in La Grande Rivière area of Nouveau-Québec; Canadian Journal of Earth Sciences, in press.

Symons, D.T.A., Osmani, I. A., and Stupavsky, M., 1982. Magnetization characteristics of the Algoman iron formation and host rocks at the Griffith Mine, Red Lake, Ontario; Ontario Geological Survey, Open File Report # , Pt. 11,

Symons, D.T.A., Quick, A. W., and Stupavsky, M., 1981. Magnetic and paleomagnetic characteristics of the Archean iron formation and host rocks at the Adams Mine, Ontario; Ontario Geological Survey, Miscellaneous Paper 98, p. 293-307.

Symons, D.T.A., and Stupavsky, M., 1979. Magnetic characteristics of the iron formation near Temagami, Ontario; Ontario Geological Survey, Miscellaneous Paper 87, p. 133-147.

Symons, D.T.A., and Stupavsky, M., 1980. Magnetic anomaly type curves and paleomagnetism of Algoman-type iron formation near Temagami, Ontario; Ontario Geological Survey, Miscellaneous Paper 93, p. 215-219.

Symons, D.T.A., Walley, D. S., and Stupavsky, M., 1980. Component magnetization of the Algoman iron formation and host rock, Moose Mountain Mine, Ontario; Ontario Geological Survey, Miscellaneous Paper 93, p. 198-214.

Telford, W. M., Geldart, L. P., Sheriff, R. E., and Keys, D. A., 1976. Applied Geophysics; Cambridge University Press, 860 p.

II. MAGNETIZATION CHARACTERISTICS OF THE ALGOMAN IRON FORMATION AND HOST ROCKS AT THE GRIFFITH MINE, RED LAKE, ONTARIO

by

D.T.A. Symons, I. A. Osmani and M. Stupavsky
Department of Geology, University of Windsor
Windsor, Ontario, N9B 3P4

ABSTRACT

The magnetic and paleomagnetic properties for the iron formation (IF) and host rocks (HR) are determined from a data base of ~1600 specimens. The Algoman-type banded iron formations have specific gravities (SG) ranging between 2.65 and 4.04 gcm^{-3}. The low-field susceptibility perpendicular to banding ($k_{\perp}$) for IF has a lognormal mean of 4.0×10^{-2} cgs cm^{-3}, so that $k_{\perp IF} \simeq 111 k_{HR}$. The regression fit equation $k_{\perp IF} = 0.116$ SG - 0.312 has a correlation coefficient of +0.85. The anisotropy of magnetic susceptibility (AMS) of $k_{\parallel} / k_{\perp}$ relative to banding is ~1.52 for the IF. The mean natural remanent magnetization (NRM) intensities for the IF and HR are 2.1×10^{-2} and 1.47×10^{-5} emu cm^{-3} which give mean Koenigsberger ratios of 0.66 and 0.051 respectively. Thus the HR contribution to the residual anomaly may be neglected. The NRM for the IF augments the calculated anomaly intensity by 14%. Combining the AMS and NRM augmenting effects with a demagnetizing factor of 2π, yields theoretical anomaly calculations for the North and South Pit deposits that are within 10% of the measured anomalies in terms of peak intensity and half-width. Utilizing alternating field, thermal and chemical demagnetization techniques along with fold and contact tests for remanence stability leads to the isolation of three Archean remanence components in both the IF and HR. The GM2 component is the oldest at ≥2.8Ga and records primary deposition or diagenesis. It may also reflect tectonic motions between the Uchi and Abitibi Belts. The GM1 component is most prevalent and records a predeformational metamorphic event at ~2.75 Ga. Its agreement with the primary components isolated in the

Archean strata of the Abitibi Belt may indicate a genetic link between the two events. The GM3 component is dominantly but not entirely post-deformational, and appears to reflect Kenoran orogenic metamorphism at ~2.6 Ga.

INTRODUCTION

This study of the Algoman-type banded iron formation (IF) deposits at the Griffith Mine is the final of four such studies. It follows the methods developed in three similar studies of the Sherman Mine near Temagami (Symons and Stupavsky 1979, 1980), the Moose Mountain Mine near Capreol (Symons et al 1980), and the Adams Mine near Kirkland Lake (Symons et al 1980) in Ontario. The objects of these studies are to provide statistically reliable data for their magnetic properties so that the magnetic anomalies over IF may be correctly interpreted and to look at their paleomagnetic properties so that their ore genesis may be better understood.

The Griffith Mine is located ~50 km south of Red Lake, Ontario (93°22'W, 50°48'N) along the west shore of Bruce Lake (Figure 1). Discovered prior to 1922 (Bruce 1924), the property was developed in the latter 1950's and early 1960's, and then brought into production by Steel Company of Canada and Pickards Mather and Company. It has a rated capacity of 1.4×10^{-6} tonnes of pellets per year.

GEOLOGY

The Griffiths IF deposit is located in the Uchi Belt of the Superior Province of The Canadian Shield. The oldest rocks are Archean metavolcanics comprised of tuff, agglomerate, rhyodacite and mafic members (Schelske 1970, Shklanka 1970)(Figure 2). Equivalent metavolcanics to the east give U-Pb zircon ages of 2959 ± 2 to 2738 ± 5 Ma (Nunes and Thurston 1980). Overlying the metavolcanics is a series of Archean metasediments of dominant greywacke composition which have been metamorphosed to quartz-plagioclase-biotite

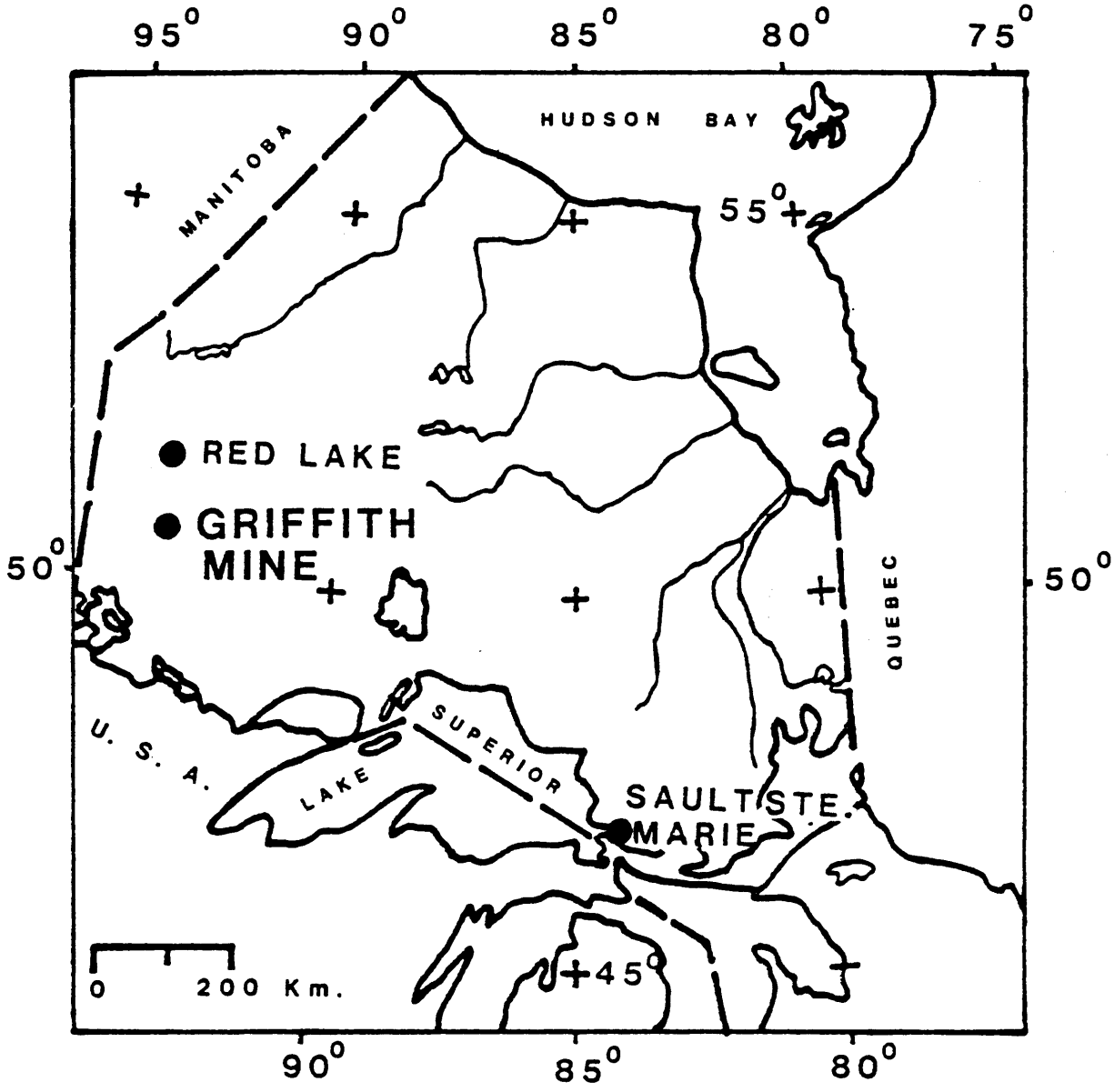


Figure 1. Location of the Griffiths Mine

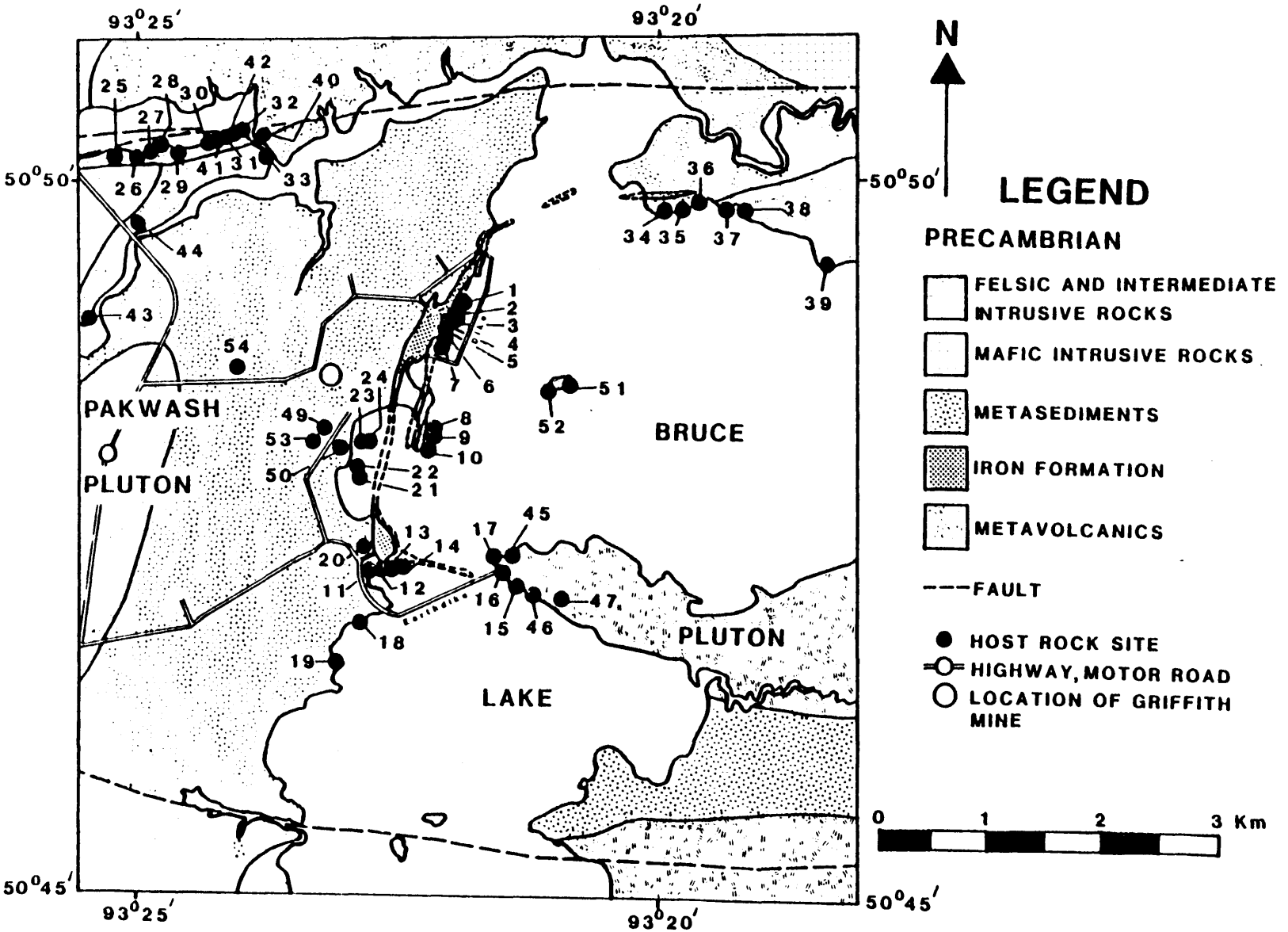


Figure 2. Geology and host rock sampling sites for the Griffith Mine.

schists with accessory minerals including magnetite, chlorite, sericite and epidote. The IF is interbedded with the metasediments. It is ~80 m thick with two members of magnetite-chert interlayered with recrystallized chert, three members of magnetite-rich biotite schistose IF, and a middle ~20 m thick member of barren biotite schist and greywacke. Medium to high grade metamorphism is indicated by the presence of accessory amphibole, actinolite, garnet and andalusite. During the Kenoran orogeny at ~2734 ± 8 Ma, the Archean strata was intensely deformed to produce tight isoclinal folds with related intense microfolding and foliations. This formed the North deposit as a syncline-anticline-syncline sequence which trends N30°E and plunges at ~30°SW and the South deposit as an overturned syncline-anticline which trends S27°E and plunges at ~60°SE.

During the Kenoran orogeny which peaked at ~2734 ± 8 Ma in the Uchi Belt, plutons were emplaced remote from the mine. Next hornblende-biotite diorite or quartz diorite plutons were emplaced which include the Bruce Pluton which forms the eastern wall rock for the deposits and extends nearly 60 km to the east, and the Pakwash Pluton which outcrops ~2 km west of the deposits. These plutons are cut by several ~50 cm wide feldspar porphyry, lamprophyre and gabbro dikes around Bruce Lake. The latter may be related to a large metagabbro pluton which is inferred from aeromagnetic data to underlie the northeastern part of the map area (ODM-GSC-1960). The latest intrusions in the area are muscovite-biotite granodiorite plutons with associated narrow aplite and pegmatite dikes found at the south end of Bruce Lake. Minor postemplacement shearing

associated with late faults is found in these rocks and in the mafic dikes which indicates a second minor period of deformation. Stockwell (1982) lists several K-Ar and Rb-Sr between ~2700 and ~2350 Ma for the plutonic rocks in the Uchi Belt which record various stages of epiKenoran cooling.

SAMPLING

At least 5 cores were drilled at each of 54 host rock (HR) sites from various units around the Griffith Mine area (Figure 2) which were oriented by solar compass. Two 2.50 cm diameter by 2.45 cm length specimens were sliced from each core giving 540 specimens. A total of 160 oriented hand samples of IF were collected from the North Pit and a further 15 from the South Pit (Figure 3). These were oriented by topographic sitings using bench mining maps. Four or more 0.95 cm right cylindrical specimens were cored from each hand sample giving a total 1070 IF specimens. Further explanation of the sampling , experimental methods and results of this study are given by Osmani (1982).

SPECIFIC GRAVITY

A modified picnometer method was used to measure the specific gravity (SG) of 452 IF specimens (~3 per sample) to an accuracy of ±0.001 g cm^{-3}. They range from barren chert at ~2.65 to ~4.04 g cm^{-3} (Figure 4) for relatively pure magnetite ore with a normal distribution having a mean of 3.36 ± 0.31 (standard deviation). Economic ore with an average of 31% total Fe has a SG of 3.60 g cm^{-3}.

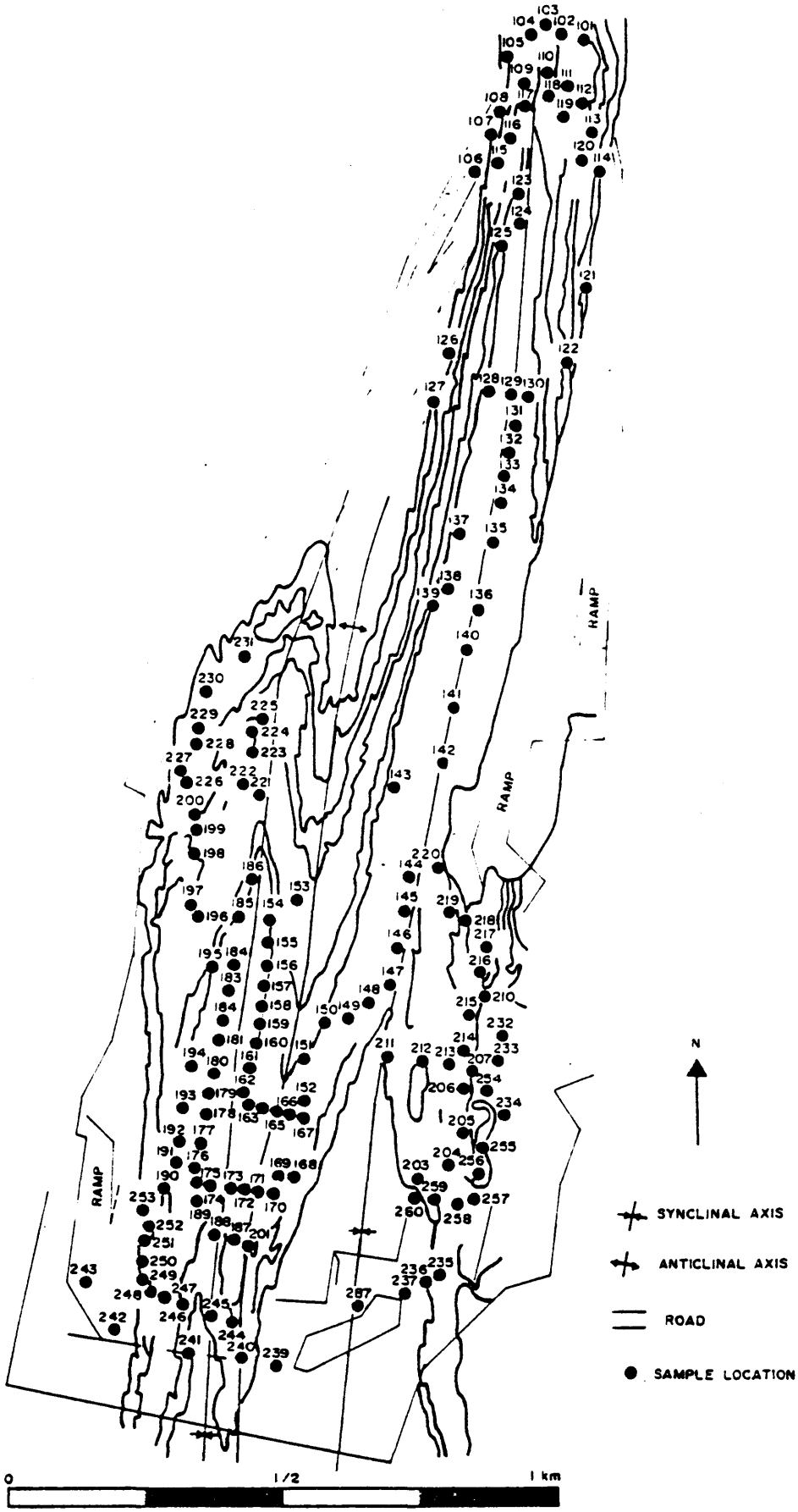
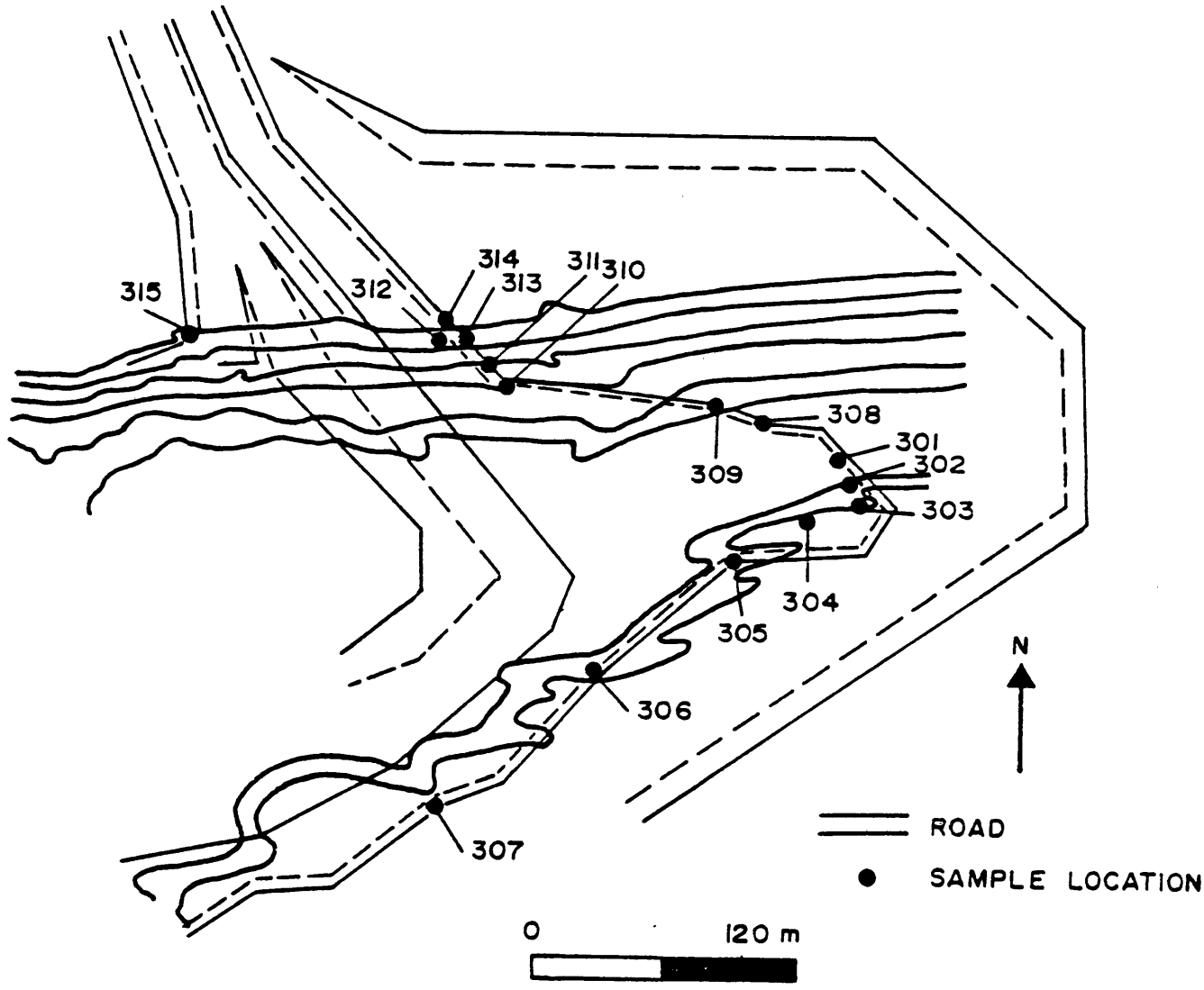


Figure 3. Iron formation sampling sites for the a) North Pit, and b) South Pit.


315
312
314
313
311
310
309
308
301
302
303
304
305
306
307
N
ROAD
SAMPLE LOCATION
0
120 m

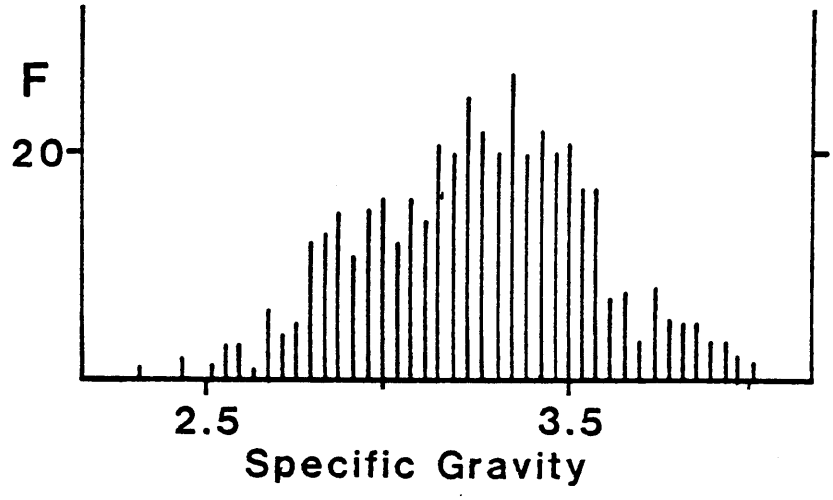


Figure 4. Histogram of IF specific gravities.

MAGNETIC SUSCEPTIBILITY

The low field magnetic susceptibility perpendicular to banding ($k_{\perp}$) was measured for 453 IF and 444 HR specimens using a toroid bridge (Christie and Symons 1959). Both data sets (Figure 5) have lognormal distributions with lognormal means of $k_{\perp IF} = 4.0 \times 10^{-2}$ and of $k_{HR} = 3.60 \times 10^{-4}$ cgs cm^{-3} respectively. Since $k_{\perp IF} \simeq 111 k_{HR}$ it is reasonable to neglect k_{HR} in calculating the IF magnetic anomaly. The regression fit between $k_{\perp IF}$ and SG is defined by $k_{\perp IF} = 0.116\ SG - 0.312$ with a high correlation co-efficient of +0.851 (Figure 6). Thus average ore for the Griffith Mine with ~27% Fe in magnetite and ~4% Fe in hematite has a $k_{\perp IF}$ value of 0.15 cgs cm^{-3}.

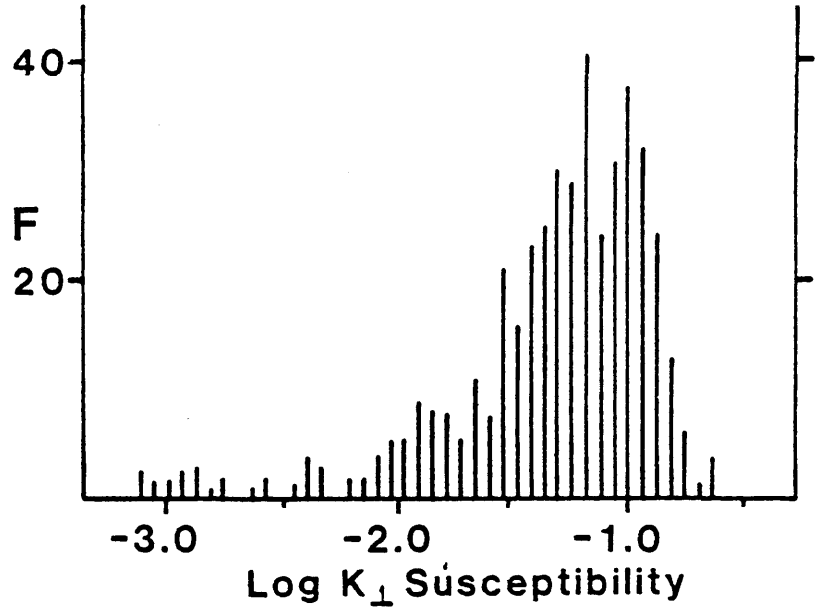
ANISOTROPY OF MAGNETIC SUSCEPTIBILITY

The low field anisotropy of magnetic susceptibility ellipsoid was determined by measuring 9 axial measurements on each of the 453 IF specimens. The axial magnitude ratios have mean values of 1.40 for k_{int}/k_{min} and of 1.50 for k_{max}/k_{min} so that the mean bedding plane susceptibility is $k_{//} = (k_{int} + k_{max})/2 = 1.45\ k_{\perp}$ (Figure 7). Because many IF specimens are confined to one band, this mean value may be somewhat lower than reality and should be regarded as a minimum. From previous studies on large cores, the true value of $k_{// IF}$ is likely ~1.52 $k_{\perp IF}$.

NATURAL REMANENCE

Iron Formation

The natural remanent magnetization (NRM) of the 453 IF specimens were measured on a Schonstedt SSM-1A spinner magnetometer (Figure 8).



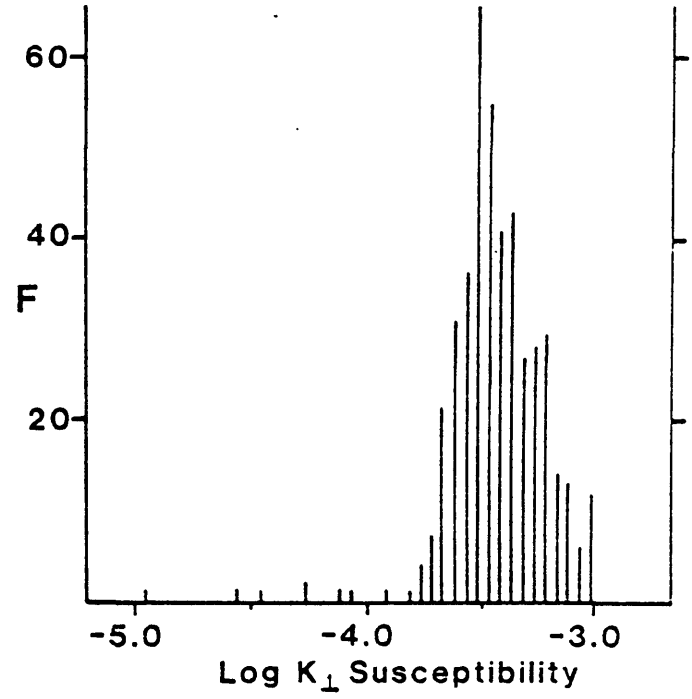


Figure 5. Histograms of $k_{\perp}$ susceptibility for a) IF, and b) HR.

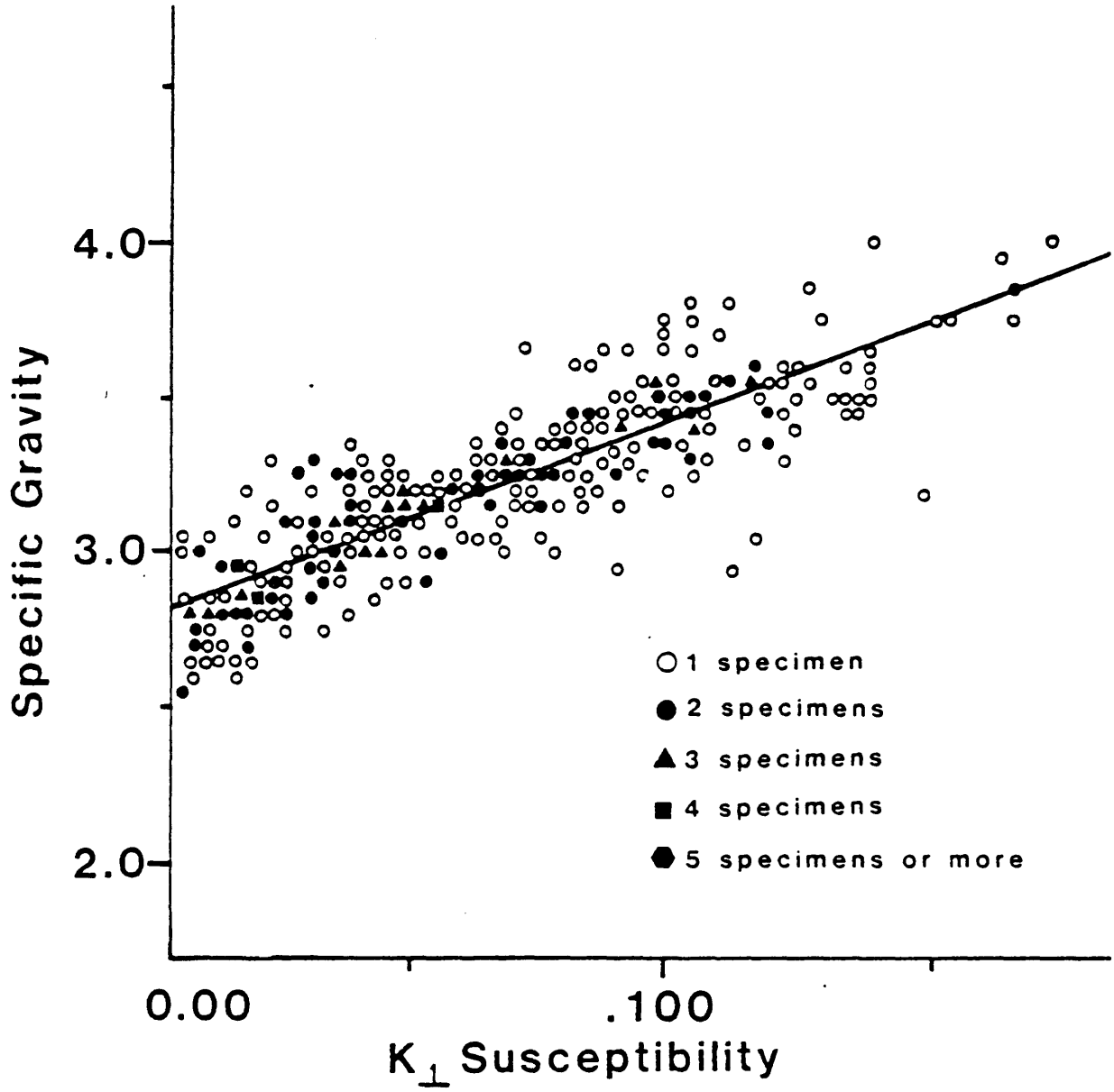
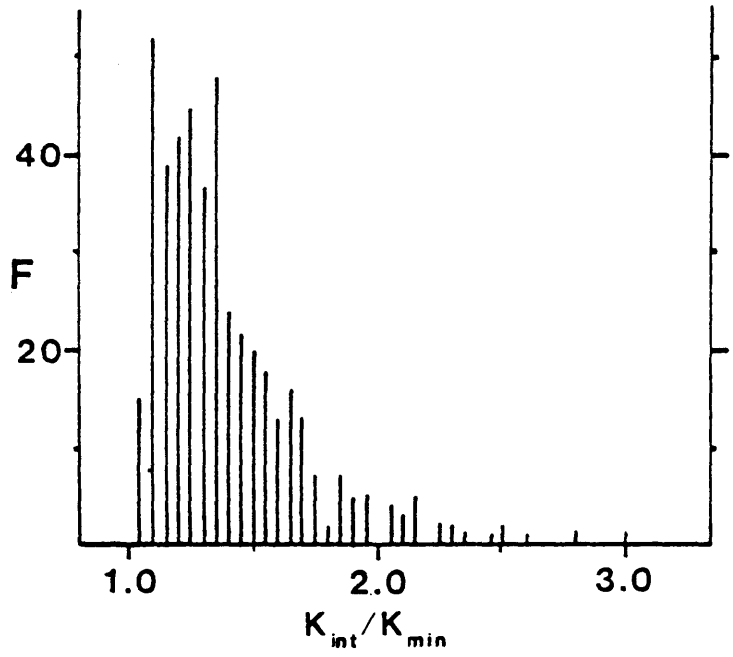


Figure 6. Regression fit between $k_\perp$ and specific gravity for IF.



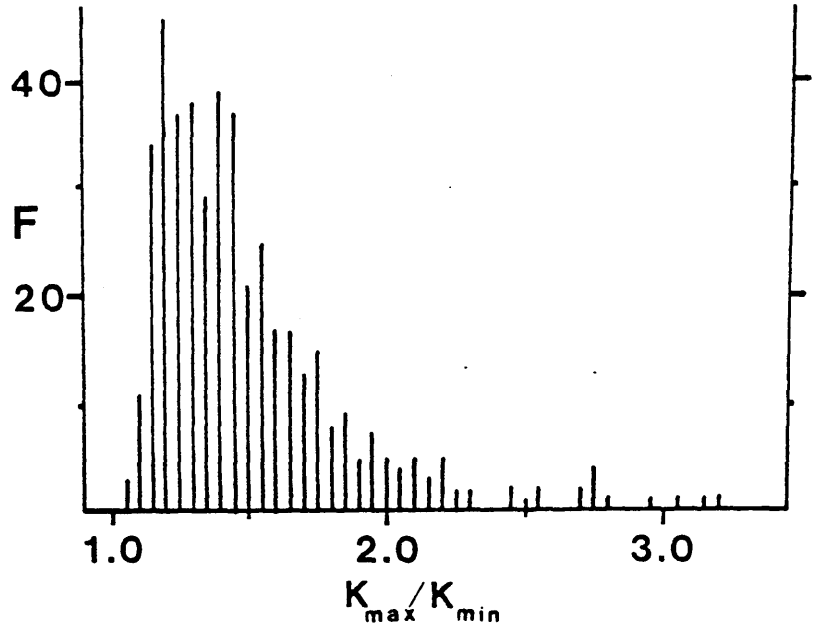
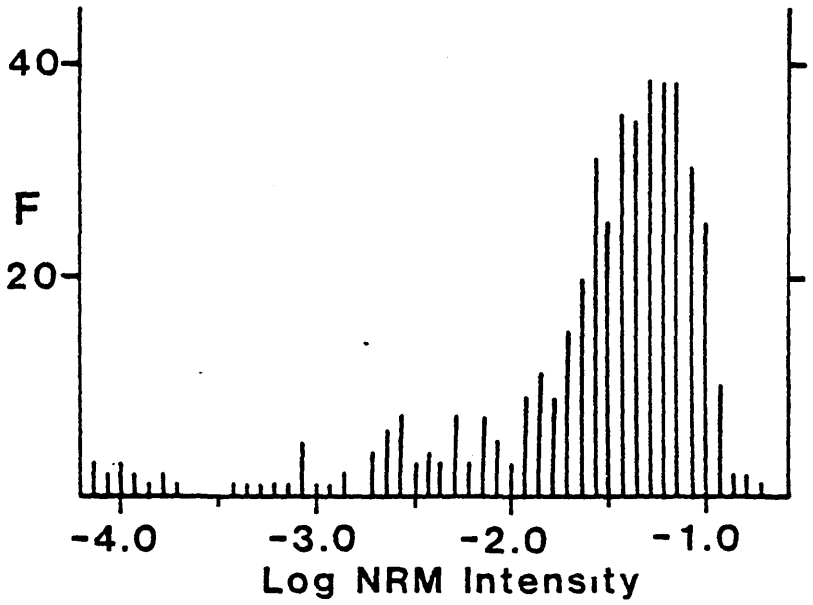


Figure 7. Histograms of a) k_{int}/k_{min}, and b) k_{max}/k_{min} for I F.



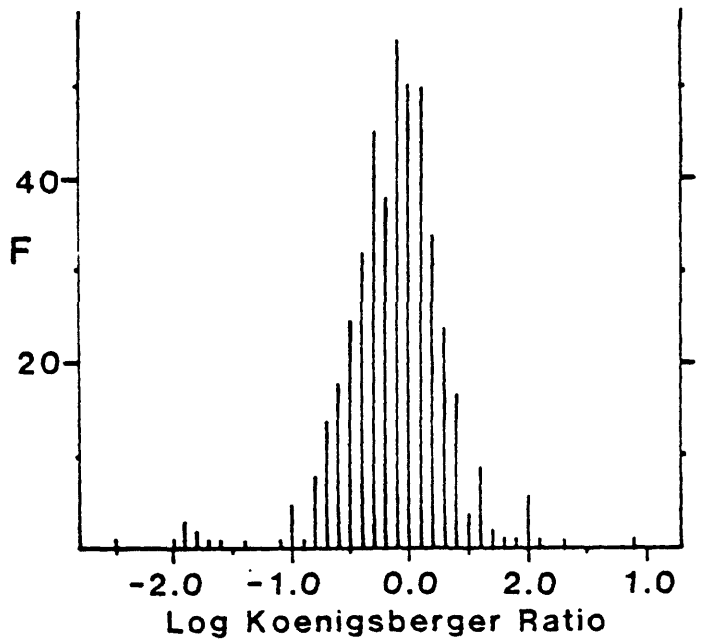


Figure 8. Histograms of a) NRM intensities, and b) Q ratios for IF.

Their NRM intensities (J_{rIF}) have a lognormal distribution with a log-normal mean of 2.1×10^{-2} emu cm^{-3}. Their Koenigsberger ratio (Q_{IF}) of remanent to induced magnetization also have a lognormal distribution with a mean of 0.66. The linear relationship of $J_{rIF} = 0.372\ k_{iIF} + 0.013$ has a good correlation coefficient of +0.55 (Figure 9). The mean NRM direction of all vectors regardless of polarity is 33.0°, 76.5° (11.5°), i.e., declination, inclination (radius of 95% confidence or α_{95} of Fisher 1953) which deviates from the present Earth's magnetic field direction of 5.5°, 78.4° by an angle (θ) of 8°. The ratio of the vector resultant (R) to the number of specimen vectors (N) gives an effective Koenigsberger ratio, $Q_e = Q(R/N)$ so that the NRM increases the induced magnetic intensity (J_i) by Q_e (cos θ) J_i to give an anomaly augmenting factor of $0.14J_i$. A storage test run on 18 representative IF specimens shows an average decrease of 17% in J_{rIF} along with a 4° change in NRM direction which indicates that the specimens have significant viscous remanent magnetization (VRM) components. A shock test was run on 20 IF specimens by striking them 20 times on an aluminum plate with their NRM vector first parallel to the EMF and then again opposite to the EMF. This test resulted in an average decrease in J_{rIF} of 44% combined with a NRM directional change of 29°. Thus dynamite blasting in the deposits may have reduced the augmenting factor by as much as ~$0.07\ J_i$, however, these VRM components are readily removed for paleomagnetic study by alternating field (AF) or thermal demagnetization.

Host Rock

The NRM intensities of 491 host rock specimens (J_{rHR}) have a lognormal distribution with a very broad spectrum and a mean of 1.47×10^{-5} emu cm^{-3}

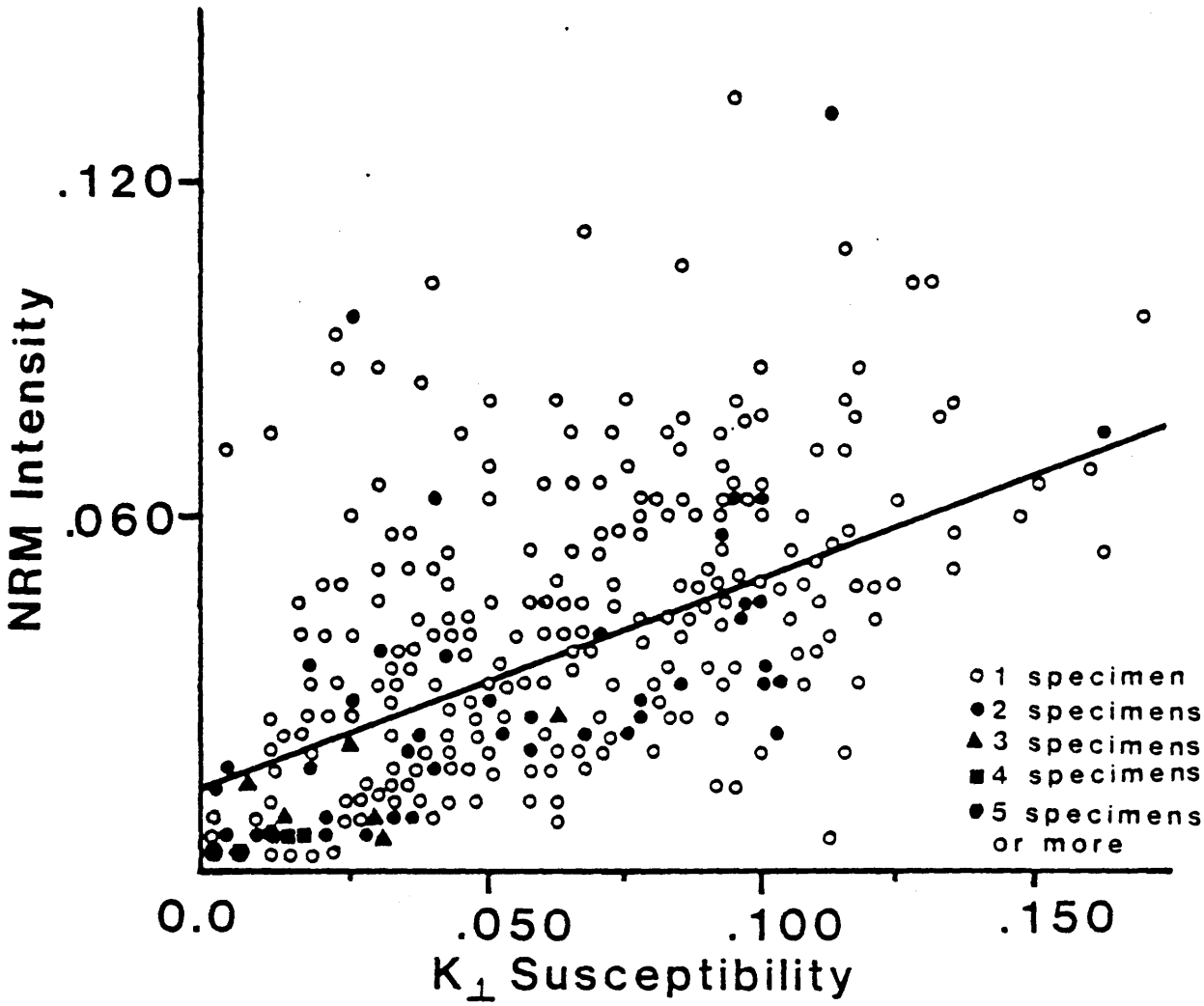


Figure 9. Regression fit between NRM intensity and $k_{\perp}$ susceptibility for IF

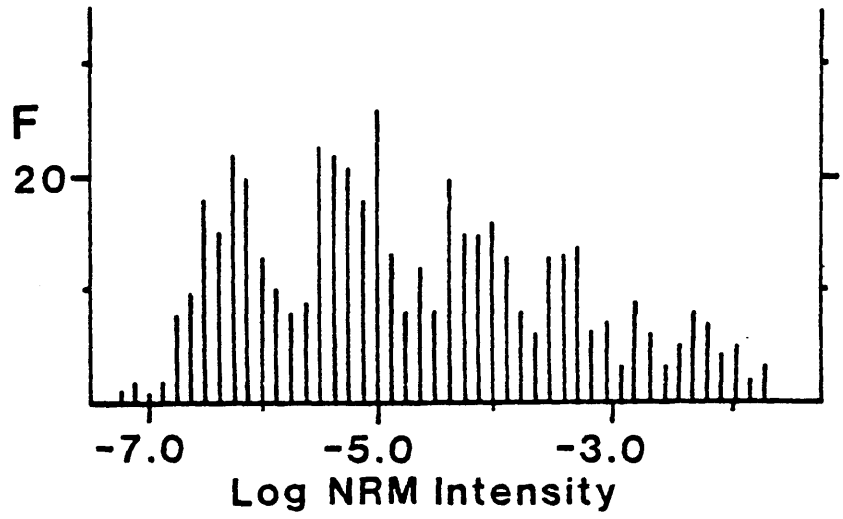
(Figure 10). Their Q_{HR} values also have a lognormal distribution with a mean of 0.051. Noting that $J_{iIF} \sim 2170\ J_{rHR}$, obviously the latter can be omitted from the magnetic anomaly calculations. A storage test on 14 HR specimens over a four week period resulted in only a 5% change in NRM intensity and a 2.5° change in NRM direction which suggests that VRM components are relatively small.

MAGNETIC MODEL

The equations for a thick tabular sheet of infinite strike and dip extent are used to calculate the theoretical magnetic anomaly. Using the north limb of the North Pit as an example (Figure 11). The input data includes: 1) the EMF direction of 5°, 78° and intensity of 60,500 γ (ODM-GSC 1960); 2) the IF strike of N30°E, dip of 85°E, and thickness of 60 m; 3) the IF $k_{\perp}$ susceptibility of 0.116 cgs cm^{-3} and $k_{//}/k_{\perp}$ ratio of 1.52; 4) the demagnetizing factor of 2π; 5) the NRM augmenting factor of 0.14 J_i; 6) the survey terrain clearance value of 147 m. The calculated anomaly has a peak value of 9200γ and a half-width of 177 m compared to the measured values of 9500γ and 145 m which represents a high degree of fit. For the South Pit equally good agreement was found with the calculated anomaly having a peak value of 11,414γ and half-width of 169 m compared to the measured values of 12,500γ and 161 m.

AF CLEANING

Alternating field (AF) step demagnetization of pilot specimens was done using a Schonstedt GSD-1 AF demagnetizer at steps of 5, 10, 15,



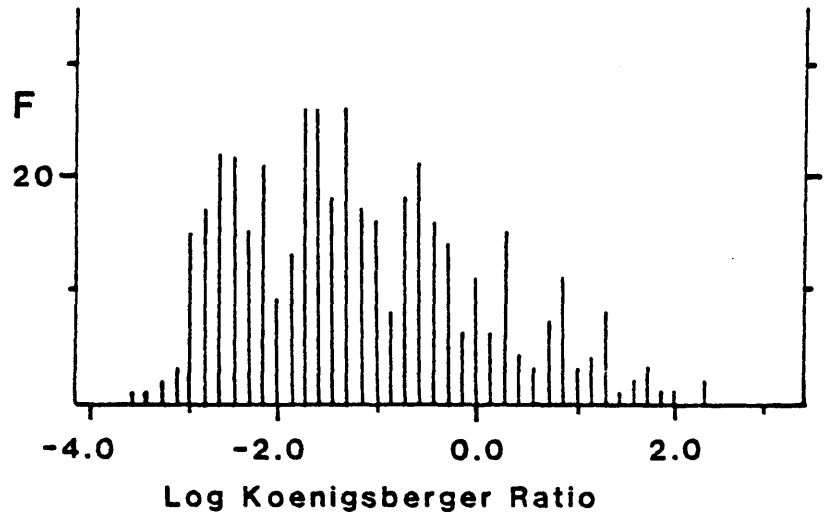


Figure 10. Histograms of a) NRM intensities, and b) Q ratios for HR.

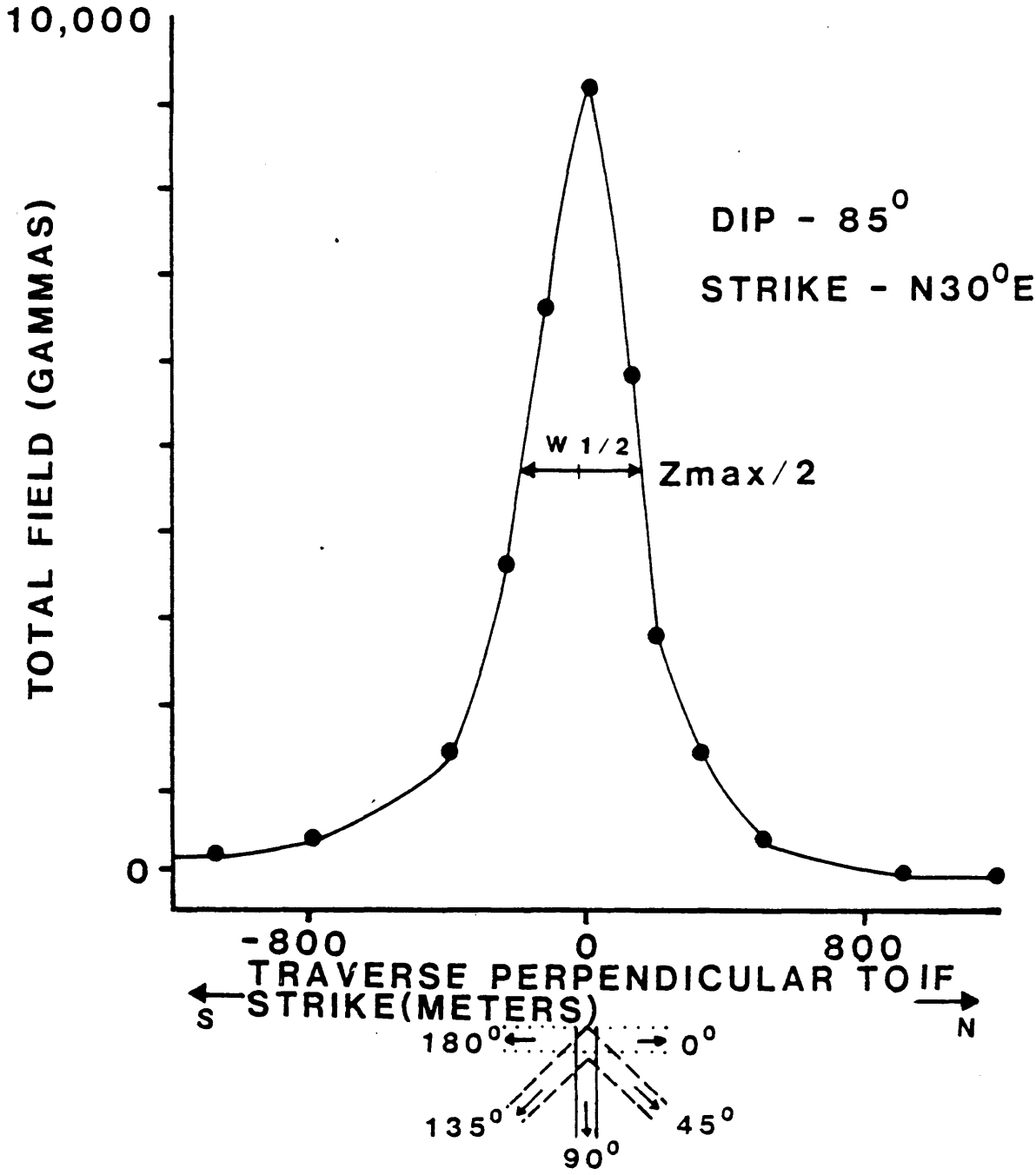
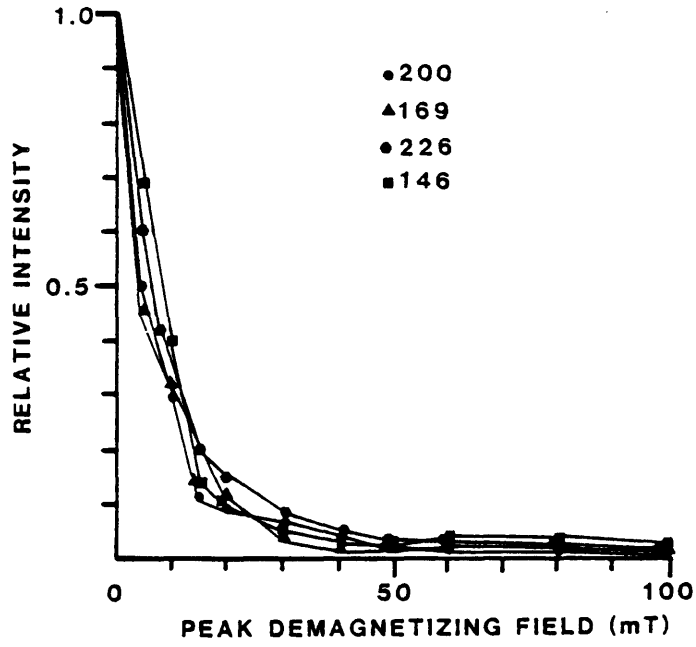


Figure 11. Calculated theoretical curve for the aeromagnetic anomaly over the north limb of the North Pit.

20, 30, 40, 50, 60, 80 and 100 mT peak fields. The paleomagnetic stability index (PSI) method of Symons and Stupavsky (1974) was used to select the optimum cleaning field for the remaining specimens as well as vector removed data and stereocontour plots. Tectonic corrections include both plunge and tilt corrections.

Iron Formation

A representative selection of 58 pilot specimens of IF were tested. More than 90% of the specimens show a rapid decay of NRM intensity down to <10% of the original intensity by ~50 mT (Figure 12). Up to 20 mT the specimens typically show rapid rates of directional motion of >100 deg T^{-1} as VRM components are rapidly removed. Between ~20 and ~40 mT, the PSI are significantly <100 deg T^{-1} as characteristic directions are isolated. Above ~50 mT the acquisition of anhysteretic remanence (ARM) components produces PSI values of >>100 deg T^{-1}. Accordingly the remaining specimens to give 4 per IF block sample were AF cleaned in fields of 30 or 40 mT. Conventional tiered statistics were first attempted whereby specimens with anomalous directions (>20° from block mean), specimens with weak remanence (<5 x 10^{-7} emu cm^{-3}), blocks with <2 reliable specimens, and block means with an α_{95}>20° were deemed unreliable. By this screening only 48 blocks, some with two component directions, or 27% of the collection yielded acceptable block mean directions. For collections of this type, modal or population level analysis (Symons and Stupavsky 1981; Van Alstine 1980) has proven more successful . In this method all vector directions for a single cleaning treatment are plotted on two stereonets (+ve inclinations on down



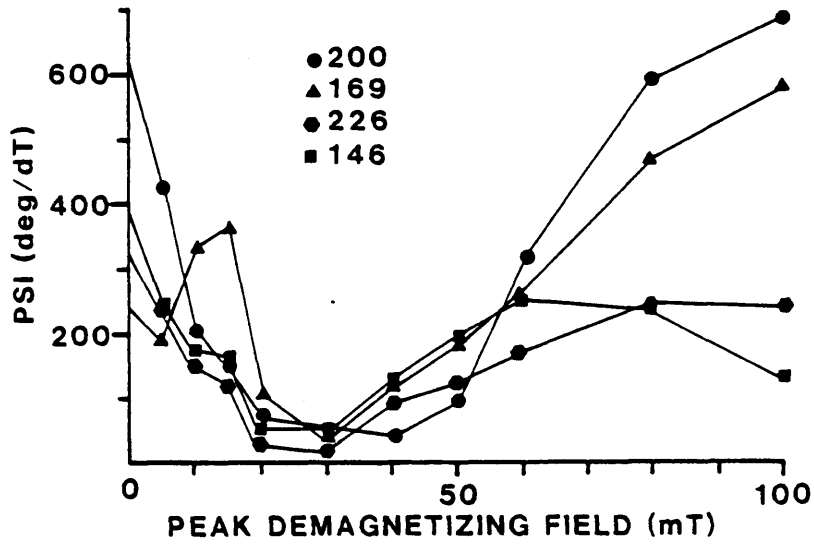
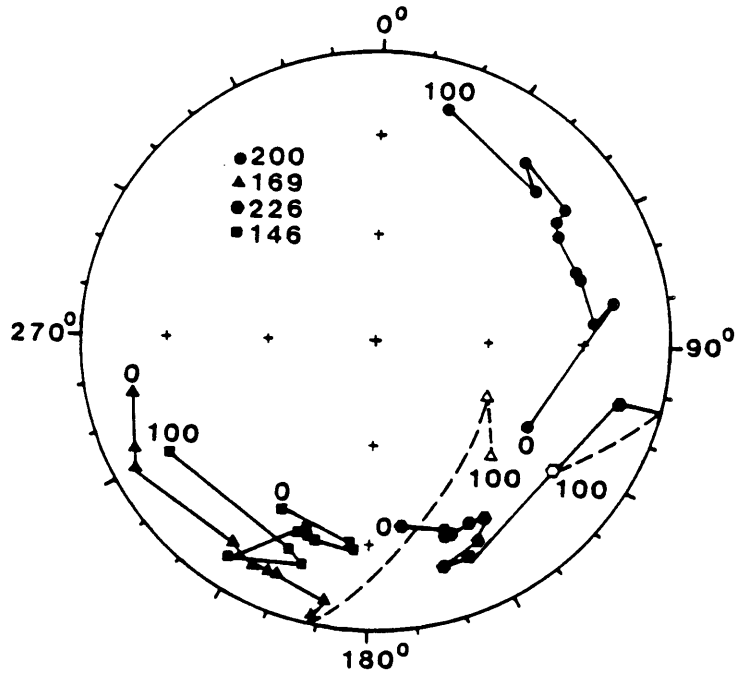


Figure 12. AF step demagnetization for example BIF specimens showing the a) decay of remanence intensity; b) change in paleomagnetic stability index (PSI); and c) directional change plotted on an equal-area stereonet with solid (open) symbols indicating down (up) directions.


0°
100
●200
▲169
●226
■146
270°
90°
0
100
0
0
100
0
100
180°

hemisphere, -ve inclinations on up hemisphere) and contoured following the method of Kamb (1959) with contour intervals in terms of standard deviations, i.e., 2 is E + 2σ or 2 standard deviations and therefore an anomaly outlined by the 2 contour has significance at >95% confidence, 4 at >99.5% confidence, etc. The plots were run both with and without tectonic correction separating out the data for the east (E1, E2, E3 anomalies) and west (W1, W2, W3 anomalies) limbs of the synclines forming the deposits and then combining the data. The results are shown in Figure 13 excluding the upper hemispheres which contain few data points. Fold tests indicate that the W3-E3 components combine to form the B component which is prefolding in origin with >>99% confidence, that the W2-E2 components combine to form the A component which is also prefolding in origin with >95% confidence, and that the W1-E1 components forming the C component are marginally postfolding in origin and appear to have been acquired when the folding was about 50% complete. The mean remanence directions and pole positions for A, B and C are given in Table 1.

HOST ROCK

One specimen per site or 54 HR specimens were AF step demagnetized. As expected they showed a wide range of NRM intensity decay patterns (Figure 14). On average they retained ~35% of their intensity at 50 mT and 20% at 100 mT. However, nearly all specimens showed small changes in NRM direction with PSI values of <50 deg T^{-1} up to ~40 mT indicating quite stable remanence as their characteristic direction was isolated.

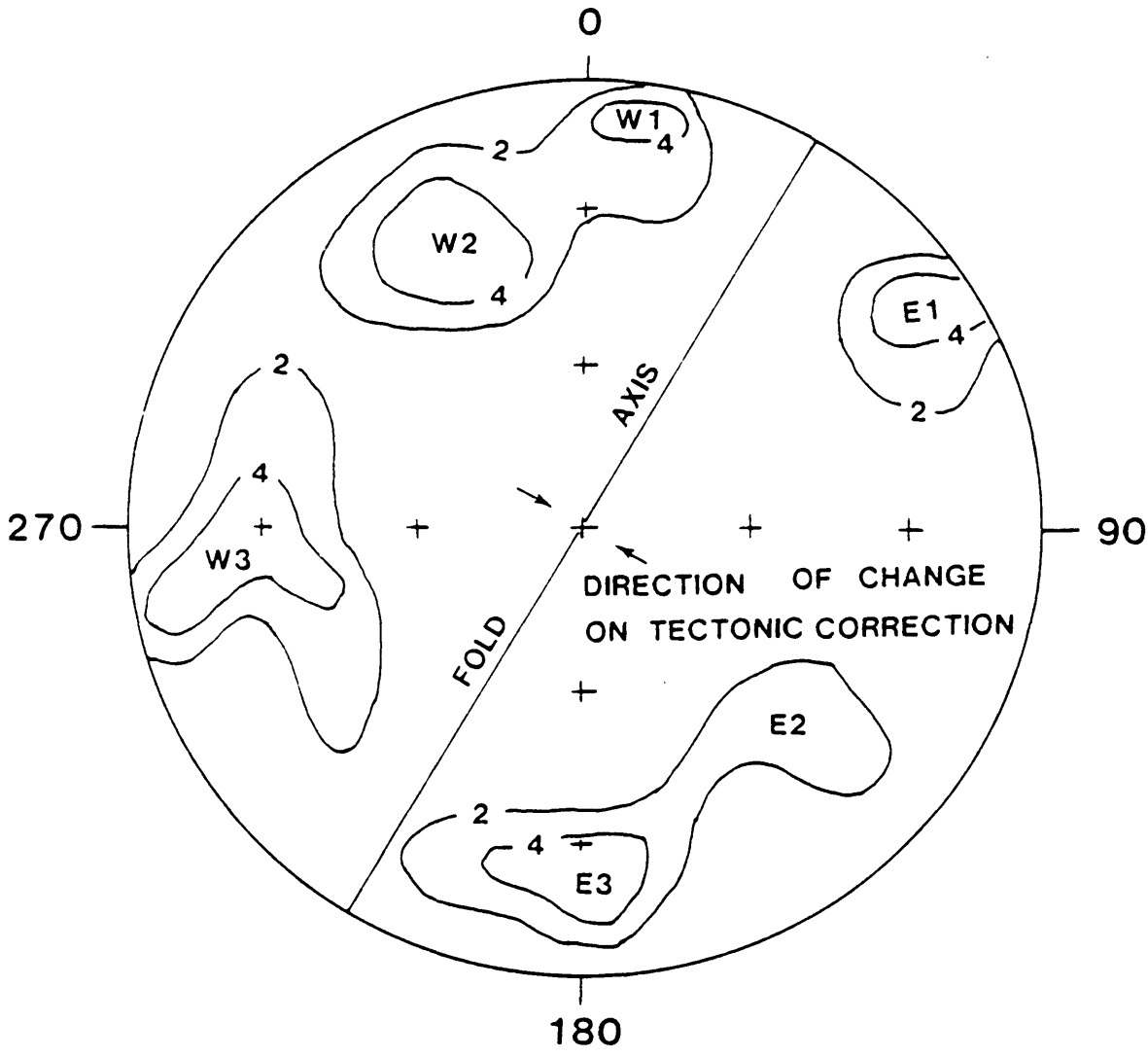


Figure 13. Modal contour plot of the IF remanence directions after AF cleaning and before tectonic correction. The contours 2 and 4 outline concentrations of >95% and >99.5% confidence respectively.

Table 1. Summary of paleomagnetic remanence components

Component	Tect. Cor.	Mean Remanence Direction					Pole Position			
		N	R	Decl. °	Incl. °	α_{95} °	Long. °W	Lat. °N	δ_p °	δ_m °
GM1-IF (B)	C	100	94.10	207.5	37.2	3.9	300.0	14.5	2.6	4.5
GM1-HR (E)	C	20	19.7	225.8	14.2	3.5	323.0	14.0	1.8	3.6
GM2-IF (A)	C	64	56.42	118.1	71.3	6.1	49.0	27.0	6.1	7.4
GM2-HR (D)	C	30	29.48	60.8	27.9	3.5	349.8	29.8	2.1	3.9
GM3-IF (C)	U-C	72	61.50	22.5	28.0	7.2	307.2	49.8	4.3	7.9
GM3-HR (F)	U-C	32	31.26	14.0	41.0	3.8	255.4	60.6	2.8	4.6

Notes: Under tectonic correction (Tect. Cor.), C indicates tectonic correction was made and U not made. N is the number of vectors averaged and R is their resultant (Fisher, 1953). The declination (Decl.), inclination (Incl). and radius of cone of 95% confidence (α_{95}) are in degrees. The pole position is given by its longitude (Long.) in degrees west (°W), latitude (Lat.) in degrees north (°N) and radii of its cone of 95% confidence (δ_p, δ_m) in degrees.

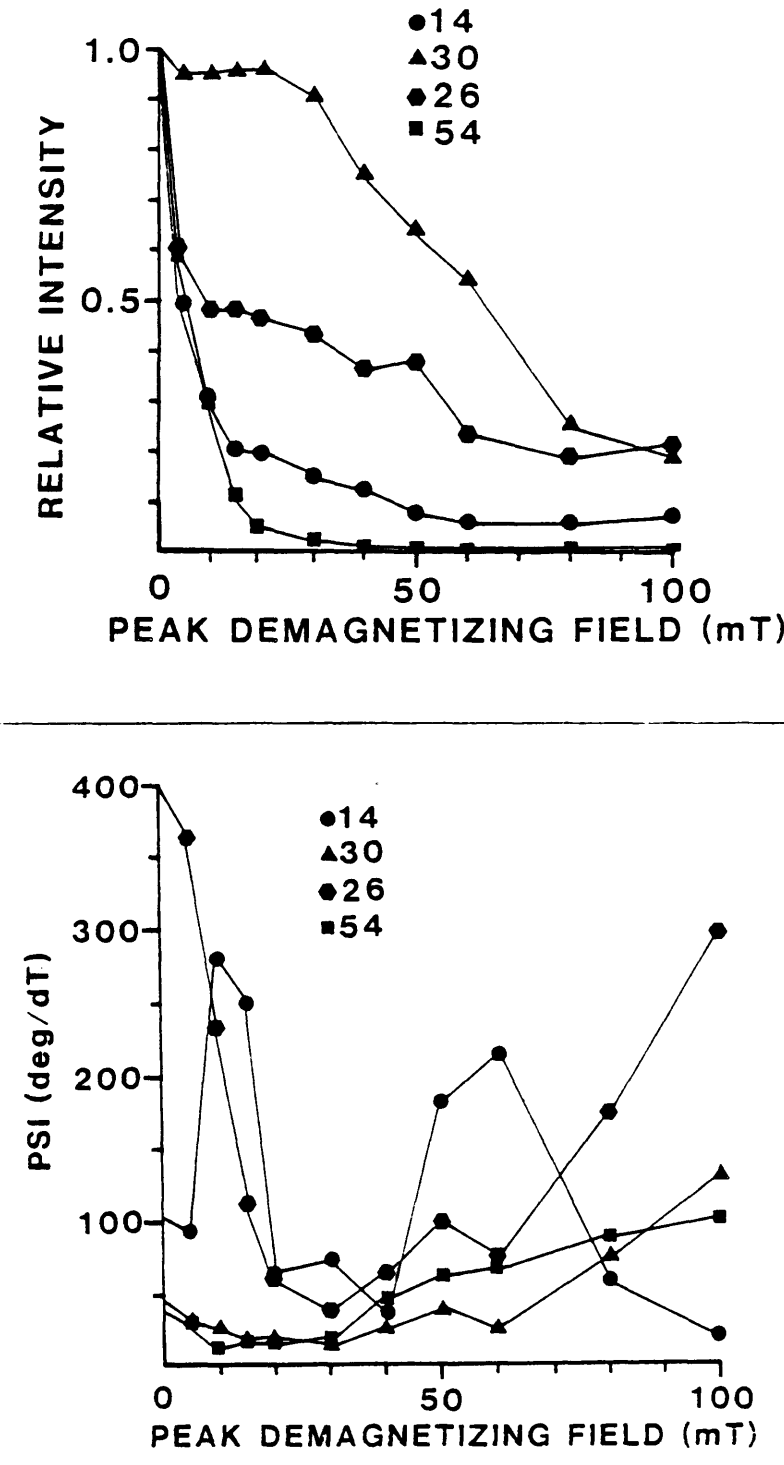
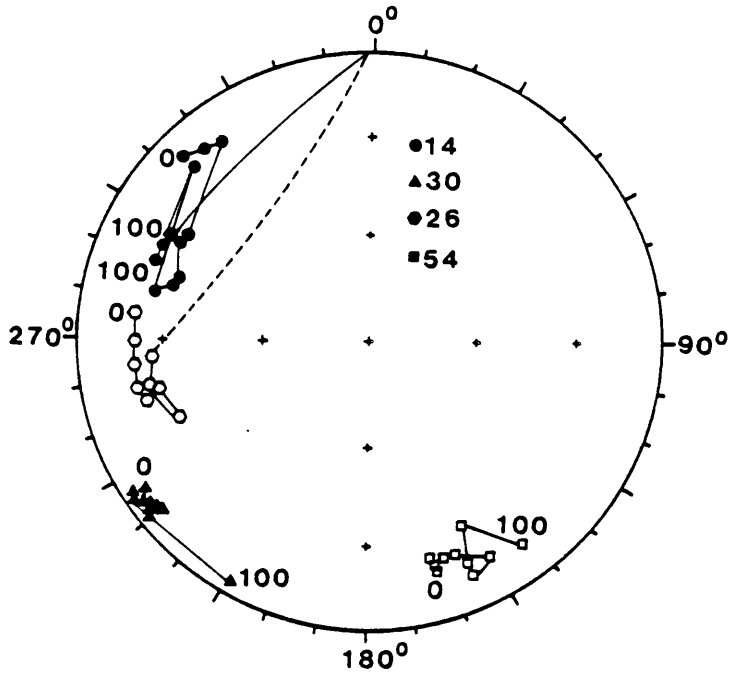


Figure 14. AF step demagnetization for example HR specimens. Conventions as in Figure 12.


0°
90°
180°
270°
●14
▲30
●26
■54
0
100
100
0
0
100
100
0

Above ~40 mT they showed an increasing scatter and correspondingly increasing PSI values as ARM components were added. Conventional tiered statistics after cleaning the remaining 8 specimens per site at their optimum field, mostly in the 20 to 40 mT range, led to the acceptance of only 19 site mean directions forming four directional clusters. Modal analysis defined these groupings far more clearly (Table 1). Fold tests on the D, E, and F components are equivocal.

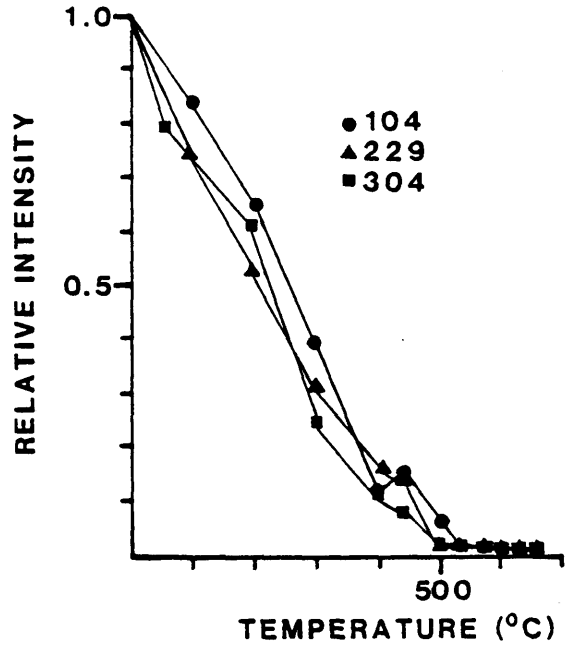
THERMAL DEMAGNETIZATION

IRON FORMATION

The 48 representative IF specimens behave quite similarly on thermal step demagnetization using steps of 100, 200, 300, 400, 450, 500, 550, 575, 600, 625 and 650°C using a Schonstedt TSD1 oven. The intensity shows a rapid near linear drop to ~15% of the NRM intensity up to ~400°C, plateaus between 400 and 500°C, and then drops to ~1% by 550°C (Figure 15). The remanence directions remain quite constant at a characteristic direction up to ~500°C and then scatter widely at higher temperatures. The isolated directions correspond mostly to the dominant W3-E3 components with some residual W2-E2 components providing clear proof that these are the most stable components. A fold test on the combined population using the 450, 500 and 550°C directions proved inconclusive due to the scatter in directions caused by the mixture of components.

HOST ROCK

Similarly one HR specimen per site, or 54 in all, was thermally step demagnetized. Six specimens disintegrated by 650°C. Most specimens show



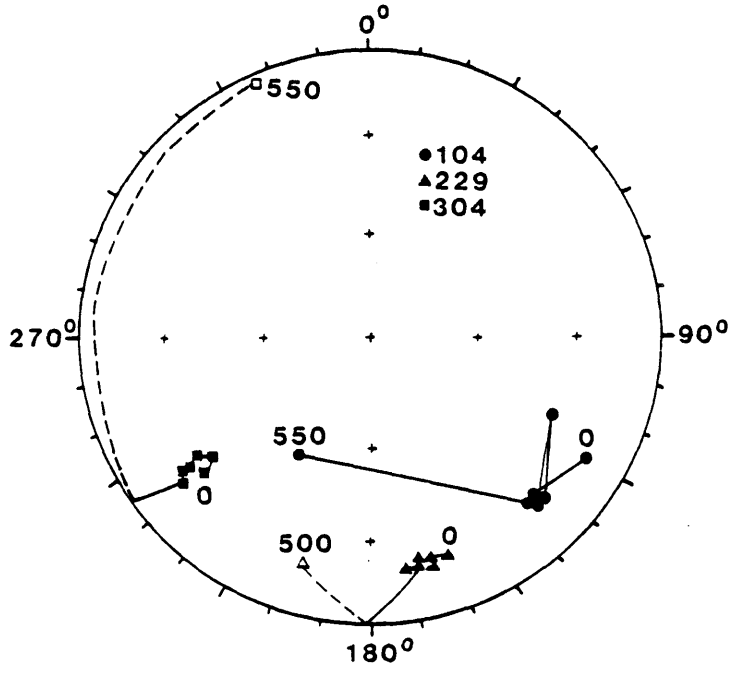


Figure 15. Thermal step demagnetization for example IF specimens showing the: a) decay of remanence intensity; and b) change in remanence direction. Conventions as in Figure 12.

a drop in NRM intensity up to ~250°C, a plateau at from 20 to 90% of their NRM intensity between ~250°C and ~500°C, and a rapid drop to ~2% of their NRM intensity by ~575°C (Figure 16), thereby indicating the presence of considerable "single domain" magnetite as the stable remanence carrier. The directions show the isolation in the 500 to 550°C range of D, E but not F remanence components that are analogous to the A and B components isolated by AF cleaning. However 54 vector directions are insufficient for full modal analysis.

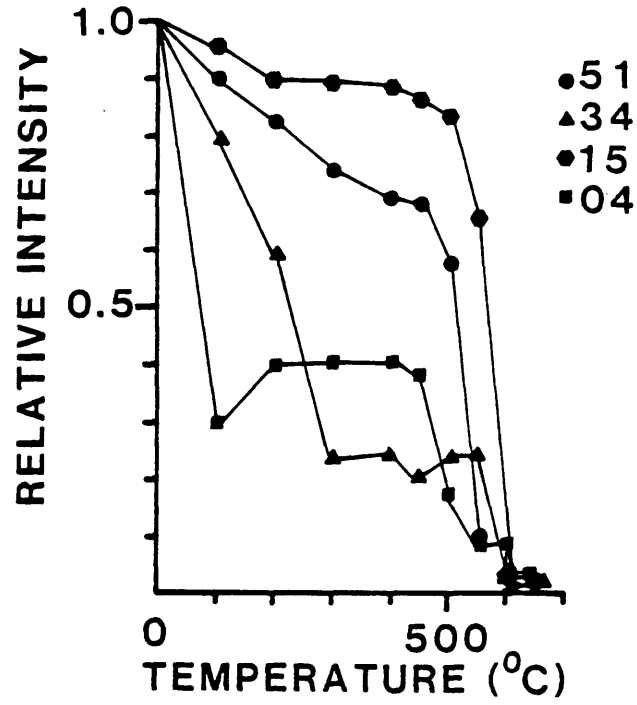
CONTACT TESTS

Two full and two partial contact tests were run on narrow (15 cm to 1 m wide) late basic dikes in contact with the HR. All tests were positive with the dike and baked zones (dike sites 7 and 32) or the dike only (dike sites 36 and 37) giving different remanence directions than the adjacent HR sites. The tests show that the remanence of the adjacent HR predates dike intrusion in the late Kenoran Orogeny.

POLE POSITIONS

The pole positions for the IF and HR components are given in Table 1 and shown in Figure 17 in relation to the apparent polar wander path of Seguin et al (1982) for the Archean time.

The IF B component is stable and definitely prefolding, and it (GM2 •) falls at ~2.7 Ga on the APW path. The HR E component is comparably stable and definitely predates dike intrusion, and it (GM2 ▲) similarly falls at ~2.7 Ga after tectonic correction. Thus both are regarded as equivalent predeformational Archean metamorphic components.



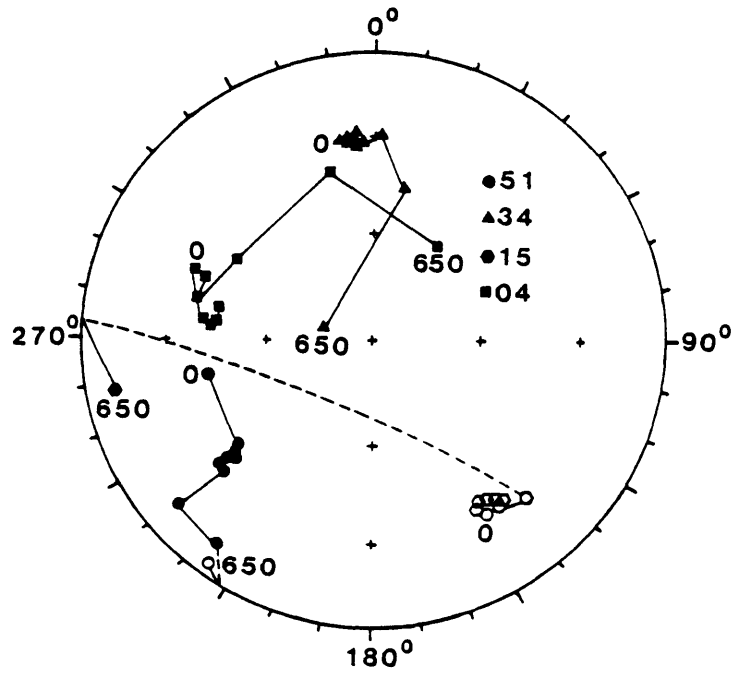


Figure 16. Thermal step demagnetization for example HR specimens.

Conventions as in Figures 12 and 15.

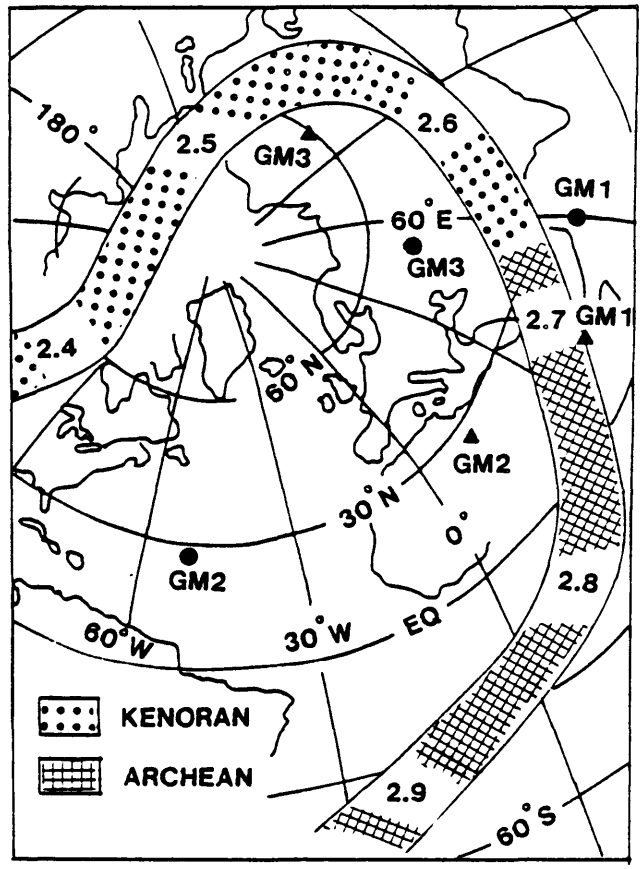


Figure 17. Apparent polar wander curve for the Archean of North America after Seguin et al (1982) showing the pole positions for the IF and HR components listed in Table 1.

The IF A component is also stable and prefolding in origin, however it (GM1 •) does not fall on any known segment of the Archean APW path. The HR D component is comparably stable and predates dike intrusion also. Its pole (GM1 ▲) also falls off the APW path, and is the closest HR pole to GM1 ▲. The difference in the poles may reflect a real difference in time between deposition of the iron formation and the enclosing HR metavolcanic and metasedimentary strata. Several possible explanations are evident for the poles not falling on the APW path. One is that the path is defined primarily by units from the Abitibi belt so that pre 2.7 Ga displacement between it and the Uchi belt is possible. Another is that the A and D components are not completely isolated so that a minor Kenoran orogeny component has biassed the poles to the north of the APW path. A third possibility is that the A and E components are ~2.9 Ga in age and thus fall on an undefined segment of the APW path.

The IF C and HR F components are probably postfolding based on the IF fold test, however, tectonic correction has little effect on the locations of the pole positions in any event. Without correction both poles (GM3 •, GM3 ▲) fall beside on the APW path at ~2.6 Ga which corresponds to the expected Kenoran Orogeny direction, and so both are interpreted to be Kenoran remagnetization overprints. This interpretation is supported by the fact that the HR sites contributing to this component come preferentially from the granodioritic Bruce Lake Pluton.

The dike poles are scattered with one close to the ~2.6 Ga position on the APW path and three closer to the ~1.7 Ga position (Irving 1979). This implies that the basic dikes may be very much younger than previously thought.

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REFERENCES

Bruce, E. L., 1924, Geology of the upper part of the English River valley; Ontario Department of Mines, Vol. 33, Pt. 4, p. 1-11.

Fisher, R. A., 1953, Dispersion on a sphere; Proceedings of the Royal Society, London, Series A, Vol. 217, p. 295-305.

Irving, E., 1979, Paleopoles and paleolatitudes of North America and speculations about displaced terrains; Canadian Journal of Earth Sciences, Vol. 16, p. 669-694.

Kamb, W. B., 1959, Ice petrofabric observations from the Blue Glacier, Washington, in relation to theory and experiment; Journal of Geophysical Research, Vol. 64, p. 1891-1909.

Nunes, P. D., and Thurston, P. C., 1980, An absolute age framework for the stratigraphy of the Uchi-Confederation Lakes metavolcanic-metasedimentary belt, District of Kenora, Ontario; Ontario Geological Survey, Miscellaneous Paper 92, p. 15-23.

O.D.M.-G.S.C., 1960, Bruce Lake, Kenora District, Ontario; Ontario Department of Mines - Geological Survey of Canada, Aeromagnetic Series, Map 861G.

Osmani, I. A., 1982, Component magnetization of the iron formation and deposits at the Griffith Mine near Red Lake, Ontario; M.Sc. Thesis, University of Windsor, 137 p.

Schelske, J. E., 1970, Mining at Griffiths Mine: Canadian Institute of Mining and Metallurgy Bulletin, November 1970, p. 1279-1282.

Seguin, M. K., Sharma, K.N.M., Woussen, G., Seymour, K. St., and Symons, D.T.A., 1982, Paleomagnetism of Archean volcanics and Proterozoic

dikes in LaGrande Rivière area of Nouveau Québec; Canadian Journal of Earth Sciences, in press.

Shklanka, R., 1970, Geology of the Bruce Lake area, District of Kenora, Ontario; Ontario Department of Mines Report 82, p. 1-25.

Stockwell, C. H., 1982, Proposals for time classification and correlation of Precambrian rocks and events in Canada and adjacent areas of the Canadian Shield - Part 1: a time classification of Precambrian rocks and events; Geological Survey of Canada Paper 80-19, 135 p.

Stupavsky, M., and Symons, D.T.A., 1982, Isolation of early Paleohelikian remanence in Grenville anorthosites of the French River area, Ontario; Canadian Journal of Earth Sciences, Vol. 19, p. 819-828.

Symons, D.T.A., Quick, A. W., and Stupavsky, M., 1981, Magnetic and paleomagnetic characteristics of the Archean iron formation and host rocks at the Adams Mine, Ontario; Ontario Geological Survey Miscellaneous Paper 98, p. 293-307.

Symons, D.T.A., and Stupavsky, M., 1974, A rational paleomagnetic stability index. Journal of Geophysical Research, Vol. 79, p. 1718-1720.

Symons, D.T.A., and Stupavsky, M., 1979, Magnetic characteristics of the iron formation near Temagami, Ontario; Ontario Geological Survey Miscellaneous Paper 87, p. 133-147.

Symons, D.T.A., and Stupavsky, M., 1980, Magnetic anomaly type curves and paleomagnetism of the Algoman-type iron formation near Temagami, Ontario; Ontario Geological Survey Miscellaneous Paper 93, p. 215-219.

Symons, D.T.A., Walley, D. S., and Stupavsky, M., 1980, Component magnetization of the Algoman iron formation and host rocks, Moose

Mountain Mine, Ontario; Ontario Geological Survey Miscellaneous Paper 93, p. 198-214.

Van Alstine, D. R., 1980, Analysis of the modes of directional data with particular reference to paleomagnetism. Geophysical Journal of the Royal Astronomical Society, Vol. 61, p. 101-113.